



PROJECT REPORT

Automated College Timetable Generator (ACTG)



KIRINYAGA UNIVERSITY

SCHOOL OF PURE AND APPLIED SCIENCES

Prepared by: DANIEL KAHIGA KAMAU (PA101/G/0356/14)

Supervised by: DR. AARON MOGENI OIRERE

A Project Report Documentation Submitted to The Department of Pure and Applied

Sciences in partial fulfillment for the award of BSc in Mathematics and Computer Science.

DECLARATION

I hereby declare that everything in this report is based on my own knowledge, understanding and has not been submitted to any institution of learning for academic awards.

This project documentation is therefore presented to Kirinyaga University, school of Pure and Applied Sciences as partial fulfillment for the award of BSc in Mathematics and Computer Science.

.....

Signature

Daniel Kahiga Kamau

.....

Date

I the undersigned do hereby certify that this is a true report for the project undertaken by the above-named student under my supervision and that it has been submitted to Kirinyaga University with my approval.

.....

Signature

Dr. Aaron Mogeni Oriere

.....

Date

ACKNOWLEDGEMENT

I would like to express my gratitude to our Almighty Father, my family for the support that they have given me this far in my school work and in this project, my lectures and my supervisor. I also thank my friends who have assisted and still support me.

ABSTRACT

An automated college timetable generator aims at reducing the time and resources spent in manual preparation of timetables. This project automates the process of allocating classrooms to different class sessions, assigning classes sessions to time periods and distributing units on offer to available lecturers while avoiding conflicts to ensure optimality of the system. This eliminates the need for faculty staff to follow a tedious and time-consuming process. More details on how the system can eliminate these shortcomings, have been explained in the problem statement. The implementation of this project was based on a genetic algorithm, where data is fetched from the database to populate functions in the algorithm and produce possible solutions. These solutions are then tested for optimality and conflicts, where only the most optimal and conflict free solution is selected as the final timetable solution. The system was coded using Java because it offers a variety of libraries useful in implementing genetic algorithms while leveraging on object-oriented programming to produce efficient code. The study followed a descriptive research design since the timetabling problem has been widely discussed previously and only a few areas are yet to be explored. The system was developed using a Rapid Application Development methodology, following the James Martin approach, for fast and efficient completion. The significance of this automation to Kirinyaga University which is the target population of the project, cannot be understated because it increases employee productivity by freeing them to concentrate on other core activities other than timetabling. Kirinyaga University though it uses a timetabling system, collisions were noted and hence it was concluded that the system is operated manually. Additionally, lecturers and students can easily access the timetables from the system. While this system achieves efficiency in timetable generation, future work could accommodate more soft constraints to allow for student and lecturer preferences such as which time periods are best for certain units. Further research could also be conducted on how such a system could be integrated with an exam timetabling system.

LIST OF FIGURES

FIGURE 1:CONCEPTUAL FRAMEWORK.....	6
FIGURE 2:REQUIREMENTS PLANNING	25
FIGURE 3: USER DESIGN.....	26
FIGURE 4: RAPID CONSTRUCTION.....	27
FIGURE 5: TRANSITION.....	27
FIGURE 6: ADMINISTRATOR USE CASE DIAGRAM	29
FIGURE 7: LECTURER USE CASE DIAGRAM	30
FIGURE 8: STUDENT USE CASE DIAGRAM	31
FIGURE 9: FLOW CHART SHOWING THE PROCESS OF TIMETABLE GENERATION	32
FIGURE 10: CONTEXT DIAGRAM.....	33
FIGURE 11: ADMINISTRATORS LOGIN	34
FIGURE 12: LECTURER OR STUDENT LOGIN	34
FIGURE 13:ADMINISTRATORS' MAIN WINDOW.....	35
FIGURE 14: LECTURERS' MAIN WINDOW	36
FIGURE 15: STUDENTS' MAIN WINDOW.....	37
FIGURE 16: ADDING TUITION DETAILS	41
FIGURE 17:ADDING ROOMS.....	41
FIGURE 18: ADDING DEPARTMENTS	42
FIGURE 19: ADDING COURSES AND UNITS	42
FIGURE 20: ADDING MEETING TIME.....	43
FIGURE 21: ENTIRE TIMETABLE VIEW AND REPORT	44
FIGURE 22: LECTURES' TIMETABLE VIEW AND REPORT	45
FIGURE 23: STUDENTS' TIMETABLE VIEW AND REPORT	46
FIGURE 24: EER DIAGRAM SHOWING DATA MODEL	47
FIGURE 25: SCHEDULE GANTT CHART.....	53

LIST OF TABLES

TABLE 1: OPERATIONALIZATION OF VARIABLES.....	6
TABLE 2:SOFTWARE REQUIREMENTS.....	28
TABLE 3: HARDWARE REQUIREMENTS	29
TABLE 4: COURSE TABLE	37
TABLE 5:TABLE DEPARTMENT	37
TABLE 6: TABLE LECTURER	38
TABLE 7: TABLE MEETING TIME.....	38
TABLE 8: TABLE ROOM.....	38
TABLE 9: TABLE TIMETABLE DATA.....	38
TABLE 10: TABLE TUITION DETAILS	39
TABLE 11: TABLE UNITS.....	39
TABLE 12: TABLE USER ACCOUNT.....	39
TABLE 14: SYSTEM BUDGET	53

TABLE OF CONTENTS

DECLARATION	i
ACKNOWLEDGEMENT	ii
ABSTRACT	iii
LIST OF FIGURES	iv
LIST OF TABLES	v
TABLE OF CONTENTS	vi
ABBREVIATIONS	ix
CHAPTER ONE	1
1.0 INTRODUCTION	1
1.1 BACKGROUND OF STUDY	1
1.2 CURRENT SYSTEM	2
1.3 PROBLEM STATEMENT.....	3
1.4 SYSTEM OBJECTIVES.....	4
1.4.0 General Objective	4
1.4.1 Specific Objectives	4
1.5 RESEARCH QUESTIONS	4
1.6 LIMITATIONS OF THE STUDY	4
1.7 JUSTIFICATION	4
1.8 SCOPE OF THE STUDY	5
1.9 CONCEPTUAL FRAMEWORK	5
CHAPTER TWO	8
LITERATURE REVIEW	8
2.0 INTRODUCTION	8
2.1 REVIEW OF THEORITICAL LITERATURE	8
2.2 CRITICAL REVIEW	13
2.3 SUMMARY OF LITERATURE	15
2.4 THEORITICAL FRAMEWORK	16
2.5 REVIEW ON EXISTING SYSTEMS.....	17
2.5.0 Time Timetable Mate	17
2.5.1 aSc Timetables	18
2.5.2 Mimosa Small School	19

2.6 RESEARCH GAP	19
2.7 REFERENCES	21
CHAPTER THREE.....	24
RESEARCH DESIGN AND METHODOLOGY	24
3.0 INTRODUCTION	24
3.1 RESEARCH DESIGN	24
3.2 SYSTEM DEVELOPMENT METHODOLOGY	24
3.2.0 Requirements Planning.....	24
3.2.1 User Design.....	25
3.2.3 Transition	27
CHAPTER FOUR	28
SYSTEM DESIGN.....	28
4.1 INTRODUCTION	28
4.2 SOFTWARE AND HARDWARE REQUIREMENTS.....	28
4.3 PROCESS	29
4.3.0 Use-Case Diagram.....	29
4.3.1 Flow chart.....	31
4.4 Input and Output.....	33
4.4.0 Context Diagram	33
4.5 User Interface	34
4.5.0 Login window	34
4.5.1 Main Window	35
4.6 Database Design	37
4.6.0 course.....	37
4.6.1 department	37
4.6.2 lecturer	38
4.6.3 meeting_time	38
4.6.4 room.....	38
4.6.5 timetable_data.....	38
4.6.6 tuition_details	39
4.6.7 unit.....	39
4.6.8 user_account.....	39
CHAPTER FIVE.....	40

SYSTEM IMPLEMENTATION AND TESTING	40
5.0 INTRODUCTON	40
5.1 PROGRAMMING LANGUAGE.....	40
5.2 TOOLS USED.....	40
5.2 IMPLEMENTATION STRATEGIES.....	40
5.3 MODULE INTEGRATION AND TESTING.....	40
5.3.0 SYSTEM TESTING	40
5.3.1 INTEGRATING THE SYSTEM WITH DATABASE	47
5.4 DATA MODEL	47
CHAPTER SIX	48
CONCLUSIONS AND RECOMMENDATIONS	48
6.0 INTRODUCTION	48
6.1 OVERALL SUMMARY	48
6.2 FUTURE DEVELOPMENTS.....	48
6.3 ACHIEVEMENTS MADE	48
6.4 CHALLENGES FACED	49
6.5 RECOMMENDATIONS	49
6.6 SUGGESTIONS FOR FUTURE WORK.....	49
6.7 CONCLUSION	49
BIBIOGRAPHY	50
References.....	50
APPENDICES	53
Schedule.....	53
Budget.....	53

ABBREVIATIONS

1. **RAD** - Rapid Application Development
2. **CSP** - Constraint Satisfaction Programming
3. **OR** - Operations Research
4. **IS** - Information System
5. **UD** - User Design
6. **RC** - Rapid Construction
7. **DBMS** - Database Management System
8. **ACTG** – Automatic Timetable Generator
9. **LP/IP** - Linear and Integer Programming
10. **RAM** – Random Access Memory
11. **HDD** – Hard Drive
12. **OCLP** - Object Oriented Constraint Logic Programming
13. **RAPS** - Resource Allocation Problem Specification
14. **CSP's** - Constraint Satisfaction Problems
15. **JRP** - Joint Requirements Planning
16. **JAD** - Joint Application Design
17. **SMS** – Short Messaging Service
18. **ACTG** – Automated College Timetable Generator
19. **GUI** – Graphical User Interface
20. **DBMS** – Database Management System
21. **SQL** – Standard Query Language

CHAPTER ONE

1.0 INTRODUCTION

This chapter will cover the background of study, problem statement, its significance, objectives, limitations and the scope of the study.

1.1 BACKGROUND OF STUDY

Timetables are time-management tools that have been in existence in society for a long time. They not only help people to plan for activities, but also schedule those activities to take place at the most appropriate times. More specifically, school timetables have for long been implemented in the education system to effectively and efficiently coordinate students, teachers, classrooms and time periods. Additionally, with different learning methods being implemented, timetables must accommodate different types of classrooms that facilitate different learning methods, such as lab sessions. The process of creating school timetables is therefore, quite time and resource intensive. Increased number of students and subjects in schools goes further to make the process complicated with more factors being considered.

Globally, teachers alongside school management are usually involved in the process of creating timetables. This activity is mostly carried out at the beginning of academic sessions, in this case semesters for colleges. It is usually a lengthy process involving a lot of compromises since considerations must be made for students, teachers and the administration. The possibility of a tutor teaching one or more subjects to more than one class poses a huge challenge to those tasked with generating timetables. Moreover, classrooms must be allocated to classes while avoiding different sessions from overlapping. These are among the challenges that timetabling presents and that must be solved.

Generating timetables has for long been a manual process that requires administrators to carry out computations aimed at balancing all the elements of a timetable. This makes it difficult to come up with a feasible solution to satisfy everyone involved. Additionally, the process is time and resource intensive, with several administrators often being involved in the process. However, with the advancement in information and communication technologies, software developers have successfully been able to automate the process of timetable generation. This has been a huge step in attempt to make the process easier and less time consuming.

The automation of timetable generation is heavily dependent on absorption of technology. As a result, the implementation of this technology is not uniform across the world. Developed countries, which generally have more resources and are usually at the forefront of not only developing but also absorbing technologies, have implemented automated timetable generators more compared to other countries. This has enabled the process to be less resource intensive whereby one person can be allocated the duty of making a timetable. These technologies, however, still need human input. Most schools in Africa still follow the manual process of making timetables. This is because most of them lack the resources to purchase or develop the software. Additionally, access to computers is still a challenge that is not sufficiently addressed in Africa. Despite these challenges, many governments are making efforts to build capacity to accommodate information technology in the education curriculum. With increased access to computer, it should be easier for schools in Africa to implement automatic timetable generators.

Information technology has been embedded in Kenya's vision to become a middle-income country. Consequently, a lot of progress has been made in increasing the accessibility of information and communication of technologies. Government has consistently made efforts to build capacity especially in the education sector. Information technology has been embedded in the curriculum which has gone a long way in encouraging innovation in the information technology sector. Tertiary institutions have led the way in this journey and have produced a skilled workforce in the information technology sector that can develop automated timetable generators. These institutions also have the economic muscle to purchase software, since most institutions have now automated several processes such as student registration. This study seeks to find out how an automated timetable generator can be implemented in such institutions, in this case; Kirinyaga University.

1.2 CURRENT SYSTEM

Normally timetable generation done manually. As we know all institutions/organizations have its own timetable, managing and maintaining these will not be difficult. Considering workload with this scheduling will make it more complex. As mentioned, when Timetable generation is being done, it should consider the maximum and minimum workload that is in a college. In those cases, timetable generation will become more complex. Also, it is a time-consuming process.

1.3 PROBLEM STATEMENT

Manual timetable generation in colleges is time consuming, risky and tedious as they are paper based. It involves a lot of paper work and manual data entry that results in scheduling errors due to data duplication and data omission also an increased number of units wastes time as the faculty staff must create a delicate balance in allocating the various units to lectures in respective classrooms. Furthermore, lengthy faculty meetings have to be held in order to submit statistics on the number of lecturers, courses and their respective units and carry out discussions on allocation of classrooms and time depending on availability of lecturers. The manual system is risky as data may be lost due to either misplacement of files or accidental fire.

The availability of classrooms complicates the generation process as faculty staff must carefully allocate various units in different courses to different classrooms while avoiding coinciding of the available classes for lectures also the number of available lecturers affects the process due to faculties often have to utilize a lecturer to teach several courses hence the faculty staff have to put into consideration the periods when the lecturer is free.

Lectures in various faculties may be assigned different time lengths posing a challenge in assigning sessions to classrooms while evading collision of the sessions. This is because different units offered by a faculty may have different time requirements, affecting the planning of classes.

The proposed system will solve the above shortcomings by;

1. Saving both time and resources (papers) as the system will generate the timetable automatically and print it. Hence minimal papers and time will be used.
2. Eliminating lengthy meetings because the system requires only one person to feed it with data and it generate the timetable automatically.
3. Ensuring all units are allocated to a classroom as it compares the capacity of the room to the number of students taking that particular unit and checks the time when the classroom is free.
4. No loss of data as data is stored in a remote database. Even if the computer malfunctions, the system can be reinstalled and data retrieved.

1.4 SYSTEM OBJECTIVES

1.4.0 General Objective

1. Come up with a system that is more robust since it uses modern java genetic algorithm and easily to use than existing systems such as aSc timetables, Time Timetable Mate and Mimosa Small School.

1.4.1 Specific Objectives

1. Generate timetables in completely automated way which saves a lot of time and effort of an institute administration.
2. Optimizing resources i.e. lecture halls.
3. Generate multiple views from timetable.
4. Provide a facility for lecturers and students to view timetable.
5. To make a timetable system generic so that it can work equally well for different colleges.
6. Eliminating the need to physically go to notice boards to get the timetables.

1.5 RESEARCH QUESTIONS

- i. How does the system optimally allocate available classrooms to class sessions?
- ii. How does the system assign classes to available time periods?
- iii. How does the system distribute units on offer to available lecturers?

1.6 LIMITATIONS OF THE STUDY

During the period of research, the researcher incurred costs in data collection and research materials which increased the costs incurred during the research period. However, it was handled through appropriate budgeting.

The respondents of the research may also be non-cooperative to give information. The researcher may give a non-disclosure agreement to cope with this.

Also, the system has some limitations and minor shortcomings when it is implemented. The system will incur the institution some costs such as paying for the SMS gateway regularly and the system needed to be maintained regularly.

1.7 JUSTIFICATION

The beneficiaries of this project will include college administrators such as the timetable coordinators since it will eliminate the time consuming and resource intensive process undergone

by university staff in generating timetables. The project seeks to automate this process and thereby equip university staff with a solution to the inconvenience that comes with the ever-changing availability of classrooms and lecturers. Other stakeholders including lecturers and students will also benefit from the project because it will provide easy access to the timetables, eliminating the need to physically go to notice boards to get the timetables. The institution will achieve higher standards in its operations in line with ISO standards and higher productivity of staff.

1.8 SCOPE OF THE STUDY

This study concentrated on timetable generation at Kirinyaga University. The project focused on an automated timetable generator compatible with all the faculties at the institution. Additionally, the project accommodated multiple user views for lecturers and students.

1.9 CONCEPTUAL FRAMEWORK

This refers to the ability to invent or formulate an idea or concept. The conceptualization phase of a project occurs in the initial design activity when the scope of the project is drafted and a list of the desired design features and requirements is created. This study will follow the following conceptual framework.

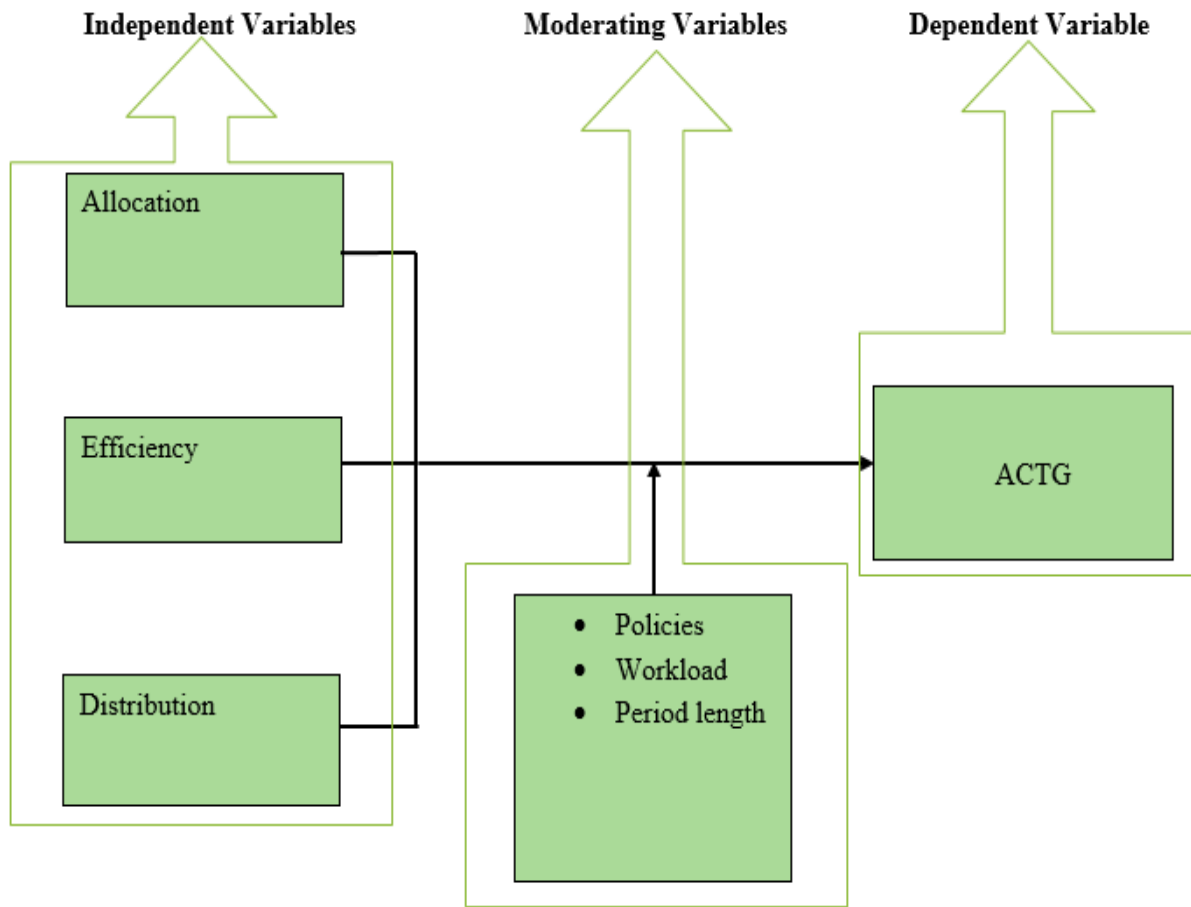


Figure 1:Conceptual Framework

Table 1: Operationalization of Variables

Independent Variables	
Variable	Explanation
Allocation	This affects how classrooms are allocated to class sessions to ensure a classroom is not allocated more than one sessions at any given time period.
Efficiency	The timetable generator must have efficiency in assigning classes to time periods without any collisions.
Distribution	This variable affects how classes are distributed to available lecturers so that a lecturer's schedule does not have any colliding sessions.

Moderating Variables	
Variable	Explanation
Policies	School policies on the maximum number of students per a given class may affect the system's performance.
Workload	This recognizes the maximum number of units assigned to a lecturer per semester as a factor that may affect the performance of this system.
Period length	The amount of time that a class session takes may affect the working of this system.

CHAPTER TWO

LITERATURE REVIEW

2.0 INTRODUCTION

This chapter presents a wide range of literature review from previous studies developed under the following headings; a review of the previously related literature, review of existing systems critical review and interpretative summary of the literature review.

2.1 REVIEW OF THEORITICAL LITERATURE

Solving the university timetabling problem is a non-polynomial-time hard (NP-hard) problem, which means that the amount of computation required for finding solutions increases exponentially with problem size. As a result, many algorithms have been suggested for use in providing the optimal solution in assigning a set of classrooms and a set of lecturers to a given set of courses. Most of the algorithms used in earlier development involve mathematical programming. However, with advancement in technology, there has been a shift towards using Artificial Intelligence to provide optimal timetable solutions by implementing constraint satisfaction algorithms.

The following literature presents a description of the various solution techniques for the university timetable problem, and how they have been previously used to meet the specific objectives of this study. These include; Integer Programming, Genetic Algorithms, Simulated Annealing, Tabu Search and Constraint Satisfaction Programming.

Integer programming is a subset of linear programming, a mathematical programming model that seeks to optimize on resource allocation. It is a technique that has been widely applied in operational research to maximize profits or minimize costs for organizations. Integer programming only uses non-negative integer values as the model variables and is therefore suitable for solving the timetable generation problem. Linear programming problems are formulated in terms of an objective functions and a set of constraints. The formulation involves representing the objective function as a linear function of the variables while the constraints are represented as linear inequalities and linear equalities. Linear programming is usually solved using the graphical method, simplex method and the transportation method.

The earliest papers devoted to the timetabling problem were mathematical in nature and based on heuristics portraying manual generation systems [1] [2]. Neither of these papers attempted to produce a realistic system, but rather they concentrated on creating a rigorous model based on the logic of the underlying problem. The development and use of heuristics then emerged in the

1960's. As the availability of computers increased during this period so did interest in utilizing their power for automated timetabling. This reliance on the mathematical “number-crunching” ability of computers prompted the proliferation in the use of LP/IP techniques for timetable generation. However, it was not long before researchers realized computational power alone would not generate satisfactory automated solutions. Indeed [3] in order to reduce specific assignment arrays. This is generally accepted as the first paper on the now classic set partitioning approach and several other papers appeared soon after which improved upon this procedure.

Simulated Annealing is a randomized local search optimization technique for finding solutions to optimization problems. It originates from physics where it was initially used to simulate the cooling of a collection of a Boltzmann distribution of atoms. Its formulation involves creating a random initial solution to the optimization problem. The simulation then proceeds to iteratively and randomly generate a neighbor of the current solution, with a different neighbor for each iteration. Subsequently, a comparison of the new solution and the current solution is done. The technique takes into account the atoms, system energy and temperature. Simulated Annealing is a problem specific technique and therefore provides a unique solution for each situation.

Three models (Graph Coloring, Set Partitioning and Simulated Annealing) were investigated by [4] to find a satisfactory solution for a multi-objective timetabling problem. The problem was expressed in terms of the optimization of a single objective function by translating some of the objectives into constraints. She discussed the advantages and disadvantages of each of these models. The significant conclusion was the proposition that given the success of an exact solution approach it is worthwhile to investigate standard models that may fit a problem even if some modification may be necessary to fit the model exactly into the given problem specific domain.

In his seminal paper (as judged by both references to in the literature and the academic community) [5] applied Simulated Annealing within a Monte Carlo scheme as an optimization technique for use in the timetable problem. He discussed the possibility of adding cost components in order to include the more complex scheduling constraints that arise in schools. The weighting of cost components allowed a component to be made more important than others. Implementing this in a parallel program, it was demonstrated that the speed of the algorithm improved along with the results. However, although the speed of the algorithm can be improved by implementation as a

parallel program the results obtained improved until overtaken by a complexity issue related to the large number of competing processors. This hardware issue was left unresolved and an issue for future research.

Trying to anticipate the eventual contradictions contained inside a partial action and state scheme [15] presented a rule based system conceived with the purpose of dealing with plan generating systems, automatic programming, timetabling or scheduling. It was claimed that this approach separately handles the semantic aspects and the temporal constraints of the system. Based on the findings of their research, although overlapping with the concepts of set partitioning, the authors had begun developing an expert system shell to assist in the timetabling process. Again, this work appears to have faded away.

In keeping with the development within the context of an expert system shell, [16] presented a paradigm RAPS for specifying resource allocation and timetable problems. They claimed that their language enabled the specification of a problem in terms of resources, activities, allocation rules and constraints. The powerful syntax of RAPS allowed it to perform allocations that are independent of its inference engine. This language, built within the expert system paradigm, enabled the solution of resource allocation problems by using experts' knowledge and heuristics. Genetic Algorithms are search algorithms based on the concepts of biological evolution. This model imitates natural selection and incorporates the concepts of reproduction, general crossover and mutation. To commence the genetic evolution, a genetic representation and a fitness function are determined. Genetic representation, called chromosome, expresses mode of the solution. The fitness function which is based on the problem is defined to measure the quality of the represented solution. It is always genetic representation dependent. Once the genetic representation and the fitness function are defined, the evolution which is an iterative process starts.

The genetic evolution has an initialization process to initialize the population formed by many individual solutions. A proportion of the existing population in current generation is then selected to breed a new generation in the selection stage. Once those individual solutions to breed a new generation are selected, reproduction takes place and the next generation population of solutions are generated through genetic operators: crossover and mutation.

By combining both traditional methods and advanced modern methods [8] discussed an automatic timetable generation method that used applications such as Graph Coloring and Genetic

Algorithms. It was claimed that their method was being implemented at some United Kingdom Universities. Although cumbersome in style, the use of hybrid approaches on this scale was successfully achieved. In the same year [8] also presented a description of the methods and techniques involving Graph Coloring and room allocation algorithms. The combination of these two methods provided a basis for a flexible and widely applicable timetable system. The problems of intractability were overcome by producing a spreadsheet-style system that the user could guide in an informed and useful way. This gave control of the search and offered the possibility of backtracking when no reasonable solution could be found, while still allowing the heuristic algorithms to do the hard work. It was concluded that although this approach did not guarantee an optimal solution, it could guarantee a solution that was feasible.

Continuing their work, the following year, [8] discussed a series of recombination operators which acted upon a direct representation of the timetable for the timetable problem. The authors compared different approaches to solving the timetable problem using evolutionary computing methods. They presented a recombination of operators and described various alternatives for incorporating heuristic knowledge into the search. This continuation of research is unusual in this area, demonstrating the effectiveness of the use of hybrid approaches.

Tabu Search is a meta-heuristic technique which tries to avoid falling into local optima by implementing a memory function which creates a special list called Tabu [17]. The search starts by seeking a local minimum and it occurs iteratively. The moves made are stored in a Tabu list so that the solutions cannot be retraced for a period of time, depending on the length of the list. This minimises the chance of cycling in the same solution, and therefore creates more chances of improvement by moving into un-explored areas of the search space. At each iteration, the move used to generate the new current solution is added to the list and the oldest element of the list is removed from this list, if it is full. The size of the Tabu list must be moderately large enough to prevent cycling, but also small enough to avoid eliminating too many potentially good solutions.

In one of the first papers in what was to become a mainstream technique in timetable solution generation, [18] investigated a number of techniques that incorporated the use of a Tabu Search. The stated aim was to minimize conflicts due to overlapping courses that involved common students, teachers, or requiring the same classroom. In addition to these classical constraints the other constraints affecting this problem that were investigated included the grouping of students

(courses taken by a large number of students have to be repeated several times during the week), compactness and precedence requirements, and geographical constraints. The model was implemented for assignment problems in general with reportedly good results.

Constraint Satisfaction involves finding values for problem variables whereby finding these values is subject to some conditions being met. These conditions are the constraints. Usually, achieving a solution that satisfies all the constraints causes conflicts, arising the need to prioritize the constraints. The constraints that have a direct impact on the objectives of the system are given high priority. A solution generated using this model is considered viable if it does not violate the constraints. Constraint Satisfaction Programming has become popular as a result of increasing application of Artificial Intelligence. It is usually implemented as a combination of logic programming and constraint solving.

A different approach asserts that many scheduling and timetabling problems can be formulated as CSP's [19]. They defined and examined basic techniques for solving CSP's, then compared them with other approaches such as integer programming, branch and bound and simulated annealing. As opposed to more developed OR methods the authors viewed CP as being a relatively new area, however they still see that there are problems for which it is competitive. With further improvements to CP methodology being anticipated, it is considered that awareness of CP as a technique for tackling combinatorial optimization problems is important for researchers in this area.

By applying OCLP, [20] described the solution for the timetable problems of privately owned colleges. The results showed that the highly-constrained problems could be solved using minimal computing resources compared to substantial man-months. It was proposed that this method (OCLP) offered many advantages, as the model presented could be applied to similar types of institutions with a minimum number of changes. Such claims are not unusual in the expert systems based papers.

[21], used a method incorporating Constraint Logic Programming to construct optimum timetables for the University of Athens Department of Informatics courses. The software platform used for the implementation was an instance of the Constraint Logic Programming class of languages, the ECL/sup i/PS/sup e/ system, and proved that the system was an appropriate vehicle for managing

the complexity of the timetable problem. A specific system was used for the timetabling procedure of a department within a university with supposedly satisfactory results.

It is suggested that the use of genetic algorithms for solving timetable planning problems is problematic due to the ambiguity in deciding the fitness function [22]. Their view is that most solutions to constraint satisfaction problems using genetic algorithms are problem-dependent and therefore difficult to apply to real-world problems. Deris [22] suggested using a hybrid algorithm consisting of a genetic algorithm and constraint-based reasoning. They tested the proposed algorithm using real data for university timetable planning. Unlike other techniques, this approach did not require modifications to the components of the genetic algorithm procedures and is therefore generic and problem independent. Consequently, the algorithm can be adapted to different kinds of problems with minimum changes and applied to most constraint-satisfaction problems.

2.2 CRITICAL REVIEW

From the above discussion, it is evident that various “traditional” optimization and artificial intelligence techniques have been applied to solve the school timetabling problem. Evolutionary algorithms and Tabu search appear to be the most popular techniques used to solve this problem. One of the drawbacks of using evolutionary algorithms is the long runtimes needed to find a solution [6]. In addition to these methods, less conventional approaches such as bee algorithms have been successfully applied to solving this problem. A new algorithm, namely, the walk up jumps down algorithm developed by Wilke and Killer (2010b) specifically for educational timetabling problems, was used to solve the German school timetabling problem. Neural networks, which are generally more suitable for performing low-level processing rather than high-level reasoning needed to solve school timetabling problems, have not only been used to successfully solve the school timetabling problem, but have also produced better results than other methods, including a genetic algorithm, Tabu search and simulated annealing, in solving five hard artificially generated school timetabling problems.

Most of the techniques presented employ some construction method to create an initial solution which is then improved by the optimization technique. These constructions methods usually allocate tuples or lessons to timetable periods with lessons and tuples that are more difficult to

schedule being allocated first. A heuristic is usually used to measure this difficulty. The two heuristics commonly used for this purpose are the number of lessons of a tuple that still have to be scheduled, tuples with more lessons are given priority, and the number of available timetable periods that the lesson or tuple can be allocated to. In the latter case a lower value indicates a tuple or lesson that is more difficult to schedule. In some cases, the construction method employs backtracking or a repair mechanism to improve the initial timetable with respect to feasibility. The use of such a construction method appears to be necessary to produce feasible and/or better-quality timetables.

As mentioned above, there have been numerous studies employing evolutionary algorithms to solve the school timetabling problem. These studies have revealed that a construction method such as that described above, without backtracking or the use of a repair mechanism, is needed to create the timetables of the initial population. Most studies using evolutionary algorithms have represented timetables directly using a two-dimensional array which the reproduction, mutation and crossover operators are applied to, to create offspring. Variations of the genetic operators either incorporating the use of hill climbing or moving tuples or teachers causing constraint violations have been necessary in some studies to produce solutions. Application of genetic operators, especially crossover, can lead to offspring, not meeting the requirements of the problem, being produced. A repair or filtering algorithm is usually employed to correct this. Elitism has also been used to improve the performance of the algorithm in some cases.

The runtimes of some of these methods are high and multiprocessing or distributed computing is needed to reduce runtimes thus enabling researchers to test the different methods more thoroughly. Some models, which can be built on, have already been developed for this purpose, [6]; [12]; [14]; [13].

Most of the research conducted has focused on implementing methods to produce feasible, good quality timetables for specific timetabling problems. However, there has been very little work [7] on the development of interactive systems that can be deployed in schools and can be easily used by administration or teaching staff to create timetables.

In all the studies described above the techniques employed have successfully solved the school timetabling problem they have been applied to. However, a criticism that can be leveled against most these studies, without in any way undermining the contribution made by them, is that the

methods have been applied to one problem only and have not been widely tested. Furthermore, in each of the three comparative studies different methodologies have produced the best results. This can again possibly be attributed to the fact that in each of the studies a single problem was used to test the performance of the different methods.

Thus, conclusions cannot be drawn from these comparative studies and the methods need to be evaluated further on additional problems.

2.3 SUMMARY OF LITERATURE

From the reviewed literature, it can be inferred that the use of heuristics to constrain the search for an initial timetable solution followed using a Genetic Algorithm is an improvement technique to arrive at a final near optimal solution. Obviously, as argued from the existing literature, an exact recipe for the optimal timetable to any general timetable problem does not currently exist.

In conclusion, it can be seen that a very large and significant literature has evolved into the timetable problem. The nature of the fundamental problem set to which timetable solution generation belongs has resulted in the development of new techniques. This has been a symbiotic relationship in that some techniques have been applied to timetable generation after being developed for the more general problem for instance graph coloring and CSP whilst other techniques developed for timetable problems have been applied back to the NPC problem set like Simulated Annealing and Tabu Search.

It is concluded from the analysis of the literature that the emerging use of hybrid solution generators, normally produced from relatively stochastic initial positions, would be iterated upon by an appropriate iterative improvement scheme to arrive at a high quality, though not necessarily optimal, solution. There is also a strong emerging trend in the reporting of University timetabling information systems. The systems reported make little or no attempt at generating timetable solutions. Rather, they are aimed at providing an IS “solution” to the administrative problem of managing the life cycle of a Timetable.

Hence, as detailed in the literature, there is a clear distinction between the technical aspects of timetable generation and management aspects of university timetable administration. In an attempt to bridge this gap, it is proposed that a system that encompasses both perspectives of timetable generation and administration be developed. There has been no successful implementation of any system that integrates these activities.

2.4 THEORITICAL FRAMEWORK

This study is based on three major theories; mutation, crossover and selection. The theories are part of the broader theory of biological evolution. The theories are the building blocks of Genetic Algorithms [8].

2.4.0 Mutation

According to [9], mutation refers to the permanent alteration of the sequence of an organism's genetic material. In genetic algorithms, it maintains genetic diversity from one generation of a possible solution to another. This concept is applied during the evolution stage of the algorithm. To commence the genetic evolution, a genetic representation and a fitness function are determined. Genetic representation, called chromosome, expresses mode of the solution. The fitness function which is based on the problem is defined to measure the quality of the represented solution. It is always genetic representation dependent. Once the genetic representation and the fitness function are defined, the evolution which is an iterative process starts. Mutation prevents the solutions from becoming too similar to each other therefore slowing or stopping evolution.

2.4.1 Crossover

This concept explains the process of taking more than one parent solutions and producing a child solution from them. It is used to vary the composition of chromosomes from one generation to another. There exist various techniques of crossover and the logic behind them is taking some part of the parent solutions and swapping them to produce a child organism. The only variation in the techniques is how the parts are chosen. The most common techniques are single-point, two-point and three parent techniques [10].

Single-point involves a single crossover point on both parents' organisms. All data beyond that point is swapped between the two parent organisms. The only variation in two-point crossover is the existence of two crossover points. Everything between the two point is swapped to produce child organisms. On the other hand, the three-parent technique involves three parent solutions chosen randomly. Each bit of the first parent is compared with the adjacent bit of the second parent. If the bits are similar it is used in the child organism, otherwise the bit from the third parent is used.

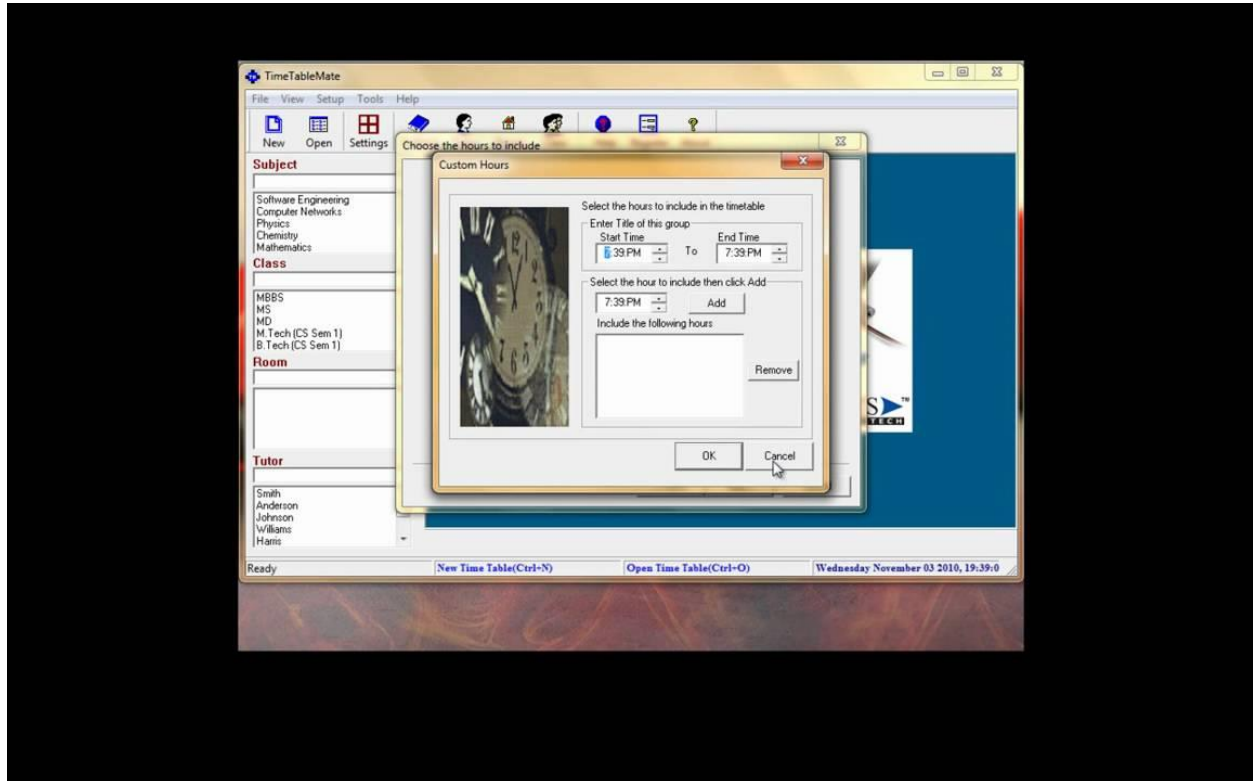
2.4.2 Selection

The process of selection is applied to facilitate mutation in genetic algorithms. A predetermined fitness function is used to evaluate individual chromosomes and each is assigned a fitness value.

The fitness value is then used to select the chromosomes based on whether they have met a predetermined criterion. The process is repeated until the individuals are enough to form a population [11].

2.5 REVIEW ON EXISTING SYSTEMS

2.5.0 Time Timetable Mate



Features

1. Automatic and interactive scheduling for tutor, class and subject
2. Clash free scheduling of tutors, classes and room
3. Low cost
4. Manual and automatic modes
5. Handy printouts of timetables
6. Options to print or save timetable in various formats.

2.5.1 aSc Timetables

The screenshot displays the aSc Timetables web application. The browser address bar shows the URL <https://testschool.edupage.org/timetable/>. The interface includes a navigation bar with icons for 'prev', 'next', 'Print', 'Subjects', 'Classes', 'Teachers', 'Classrooms', 'Students', 'Summary timetables', and 'Published Timetable 01'. The main content area shows a timetable for 'greenfield flower school' titled '5.A'. The timetable is a grid with days of the week (Mo, Tu, We, Th, Fr) as rows and time slots (1 to 7) as columns. Each cell contains a subject name (e.g., En, Ma, Pa, Sp, Hi, Ge, Na, Mu) and a color-coded background. A large red watermark text 'New timetable viewer for web browsers' is overlaid diagonally across the timetable. At the bottom left, it says 'Validity: 1/9/2016-30/6/2017' and 'aSc Timetables - school scheduling software'. At the bottom right, it says 'aSc Timetables Online'.

	1 8:05 - 8:45	2 9:00 - 9:45	3 10:00 - 10:45	4 11:00 - 11:45	5 12:00 - 12:45	6 13:00 - 13:45	7 14:00 - 14:45
Mo	En	Ma	Pa	Sp	Hi	Ma	
Tu	Sp	G	Hi	En	Ge	Ma	
We	En	Ph	Ma	Na	Sp		
Th	Ma	Ge	Sp		G		
Fr	Na		Ph	Ma		Mu	

Features

1. No installation is need.
2. Online timetable generation.
3. Online storage – backup.
4. Feature for lectures only to view timetables.

2.5.2 Mimosa Small School

The screenshot displays the Mimosa Small School software interface. The top menu bar includes File, Edit, View, Tools, Options, Window, and Help. Below the menu is a toolbar with icons for New, Open, Save, Publish, Resources, Events, Weeks, Timetables, Colours, Options, Optimise, Health check, and Web. The main window is divided into two panes. The left pane lists courses with columns for #, CODE, NAME, and a color-coded status. The right pane shows a timetable grid with columns for X, Mo, Tu, We, Th, and Fr. A pop-up window is visible over the timetable grid, showing a detailed view of a specific course (RUS241) with its own timetable grid.

#	CODE	NAME	* PLANNED	ALLOCATED	SCHEDULED	LINKS	RESOURCES
1	MOD187	Modelling and Moulding	a	23	23	23	8x 4G+2L+2R
2	MUL873	Multi-Cultural Work	f	46	46	46	6x 2G+2L+2R
3	EVE741	Everyday English	b	92	92	92	6x 2G+2L+2R
4	YOU280	Youth Education and Guidance	f	46	46	46	5x G+2L+2R
5	TRA381	Transport and Insurance Laws	b	60	60	60	5x 2G+2L+R
6	TRA360	Transport and Insurance Laws	b	30	30	30	5x G+2L+2R
7	SWE492	Swedish for Marketing	m	75	75	75	5x G+2L+2R
8	RUS732	Russian for Beginners	b	60	60	60	5x G+2L+2R
9	RUS241	Russian for Businessmen I	m	92	92	92	5x G+2L+2R
10	PRO71	Probation and Aftercare					
11	PRA391	Practical Training					
12	PLA463	Planning of Renovation					
13	MET95	Methods of the Aged					
14	MEE283	Meetings and Negotiations in C					
15	MAT70	Mathematics II					
16	LEA893	Leadership					
17	ITA652	Italian for Beginners II	m	60	60	60	
18	INT826	Intoxicants and Dependence	a	92	92	92	
19	INT622	International Economics	a	38	38	38	
20	INT107	Introduction to Corporate Finance	a	30	30	30	
21	GUI305	Guidance of Open Social Care	m	60	60	60	
22	GER951	German Structure	b	92	92	92	
23	FRE395	French for Beginners I	a	92	92	92	
24	FOL431	Follow-up Course in Russian	b	90	90	90	
25	EVE665	Everyday French	b	76	76	76	
26	ELE479	Electrical Machines	f	46	46	46	
27	BUS186	Business Basics in German	m	76	76	76	
28	ADV733	Advanced Spreadsheet in Business	f	60	60	60	
29	ADV225	Advanced English	m	63	63	63	
30	VAL26	Value Bucking	m	46	46	46	
31	SWE995	Swedish	a	60	60	60	
32	SPE351	Specialised Project	b	30	30	30	
33	SHI794	Ship Systems	f	30	30	30	
34	REM799	Remote Sensing	b	92	92	92	
35	INT792	Intercultural Communication	a	30	30	30	
36	INT309	Intercultural Communication	b	30	30	30	
37	EXT547	Extension Course in Statistics	a	138	138	138	
38	EXH544	Exhibition Design	f	75	75	75	
39	ENG824	Engineering Materials	f	60	60	60	
40	ELE837	Electrical Materials	b	92	92	92	
41	DES984	Design Management	m	92	92	92	
42	CRI432	Criminal Policy	m	63	63	63	

Features

1. Timetables and other reports can be printed and published as webpages for your students, staff, clients, or any others interested or entitled.

2.6 RESEARCH GAP

As evidenced by the reviewed literature, existing studies on university timetabling seek to either handle the technical aspects of generating a timetable solution or providing an information system to support administrative functions. However, there exists a need to provide a system that will not only output an optimal university timetable but also provide a timetable information system that will support decision making by all stakeholders of the institution. This study seeks to fill this gap.

Also, by reviewing the existing similar systems it was found that there still exists a gap in terms system efficiency and complexity of user interface. Comparing Time Timetable Mate with ACTG, it was found that Time Table Mate is not friendly to both tutors and students as they have to wait for the system administrator to generate timetables and present handy printouts to them. ACTG is

friendly as it offers a portal to both students and lectures to filter, view and print timetables for themselves.

By Comparing the aSc timetables with ACTG, aSc timetables is an online system hence it uses a scripted programming language for generation while ACTG uses a compiled language (Java 8). Compiled languages are considered more efficient in terms of execution time as they are precompiled so any trivial algorithms are executed faster. In addition, if the internet connection fails, then the process is interrupted. ACTG is a native application and no internet connection is needed to generate timetable.

By Comparing the Mimosa small school with ACTG, the mimosa small school is also web based and hence its backend code uses a scripted language as well hence it's not efficient as compared to ACTG. Also if the internet connection fails, then the process is interrupted. ACTG is a native application and no internet connection is needed to generate timetable.

The one common feature that all the systems studied above lack is that they not able to send immediate notifications to the respective users that a timetable has been generated. The ACTG fills this gap as it uses both SMS and Email gateways to send notifications to all users concurrently.

2.7 REFERENCES

1. Kuhn HW, 1955. The Hungarian method for the assignment problem. Naval res Logist quart.
2. Haynes JG, 1959. The conference scheduling problem. Operations research society of America 16th annual meeting Pasadena California.
3. Gotlieb C C, 1963. The construction of class-teacher timetables. Proceedings of IFIP congress 1962 Amsterdam.
4. Dowsland, 1990. A Timetabling Problem in Which Clashes Are Inevitable. Journal-of-the-Operational-Research-Society. 2179
5. Abramson, D. 1991. Constructing School Timetables Using Simulated Annealing: Sequential and Parallel Algorithms Management-Science.
6. Abramson, D., & Abela, J. (1991). A parallel genetic algorithm for the solving the school timetabling problem. In *Proceedings of the fifteenth Australian conference: division of information technology, C.S.I.R.O.* (pp. 1–11).
7. Moschopoulos, C. N., Alexakos, C. E., Dosi, C., Beligiannis, G. N., & Likothanassis, S. D. (2009). A user friendly evolutionary tool for high-school timetabling. *Studies in Computational Intelligence*, 166, 149–162.
8. Burke, E. K., Elliman, D. G., and Weare, R. F 1994-1995. The automation of the timetabling process in higher education Journal of Educational Technology Systems
9. Mitchell, M. (1996). *An introduction to genetic algorithms*. Cambridge, Mass: MIT Press.
10. Michalewicz, Z. (1992). *Genetic algorithms + data structures =: Evolution programs*. Berlin: Springer-Verlag.
11. Gwiazda, T. D. (2006). *Crossover for single-objective numerical optimization problems*. Łomianki: TOMASZGWIAZDA E-BOOKS.

12. Kingston, J. H. (2007). Hierarchical timetable construction. In E. Burke & H. Rudova (Eds.), *Lecture notes in computer science: Vol. 3867. PATAT VI* (pp. 294–307).
13. Kingston, J. H. (2010). Solving the general high school timetabling problem. In *Proceedings of the 8th international conference on the practice and theory of automated timetabling (PATAT 2010)* (pp. 517–518).
14. Srndic, N., Dervisevic, M., Pandzo, E., & Konjicija, S. (2009). The application of a parallel genetic algorithm to timetabling of elementary school classes: a coarse grained approach. In *Proceedings of ICAT 2009–2009 22nd international symposium on information, communication and automation technologies* (pp.1–5). New York: IEEE Press.
15. Gudes-E, Kuflik-T, and Meisels-A, 1990. On resource allocation by an expert system Engineering-Applications-of-Artificial-Intelligence.
16. Solotorevsky-G, Gudes-E, and Meisels-A, 1991. Specifying resource allocation and time-tabling problems using a rule-based language. Proceedings. The Fifth Israel Conference on Computer Systems and Software Engineering (Cat. No. 91TH0349-1). IEEE Comput. Soc. Press, Los Alamitos, CA, USA; 1991; xiv+190 pp.
17. Reeves C, 1993. Modern Heuristic Techniques for Combinatorial Problems, Blackwell Scientific Publications, Oxford.
18. Hertz-A, 1991. Tabu search for large scale timetabling problems. European Journal-of-Operational-Research.
19. Brailsford, S. C., Potts, C. N., and Smith, B. M. 1999. Constraint satisfaction problems: Algorithms and applications. European Journal of Operational Research 119.
20. Deris-S, Omatu-S, and Ohta-H, 1995. Course scheduling and timetabling by constraint-based reasoning. Bulletin-of-University-of-Osaka-Prefecture, -Series A.

21. Frangouli-H, Harmandas-V, and Stamatopoulos-P, 1995. UTSE: construction of optimum timetables for university courses-a CLP based approach. Practical Application of PROLOG. The Proceedings of the Third International Conference on the Practical Application of Prolog. Alinmead Software, Blackpool, UK; 1995; 719 pp.
22. Deris S, Omatu S Hiroshi Ohta, Saada, P., 1999. Incorporating constraint propagation in genetic algorithm for university timetable planning. Engineering Applications of Artificial Intelligence (12) pp. 241-253.

CHAPTER THREE

RESEARCH DESIGN AND METHODOLOGY

3.0 INTRODUCTION

This chapter focuses on the research methodology that will be used to conduct the study. It describes the research design, and the system development methodology to be used.

3.1 RESEARCH DESIGN

The study will follow a descriptive research design. Descriptive research is a process of collecting data to test hypothesis or to answer questions concerning the status of the subject of the study. The area of timetable generation has been widely tackled and only a few aspects are yet to be discovered thus the relevance of the descriptive design. The study will therefore be aimed at revealing the characteristics of the phenomena. Additionally, the study will be a case study, as a secondary research design since it targets a specific organization therefore allowing for in-depth study of the organization.

3.2 SYSTEM DEVELOPMENT METHODOLOGY

The system will be developed using a RAD methodology, specifically following an approach described by James Martin in 1991. The methodology has the following four stages: Requirements Planning, User Design, Rapid Construction, Transition.

3.2.0 Requirements Planning

The RP stage begins with a meeting or series of meetings, between the customers and IS management. These meetings initiate the development process by establishing a mutual understanding of the objectives of the development project in general and the system in particular. Business executives, end users and IS professionals, then, take part in JRP workshops which progress through a structured set of steps. During this stage, an outline of the system area and definition of the system scope are developed. The figure below illustrates the stage.

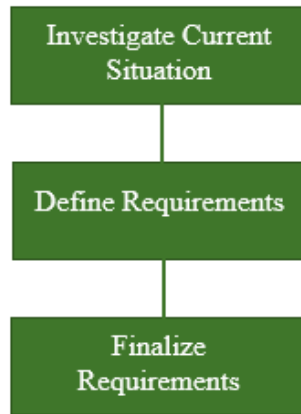


Figure 2: Requirements Planning

3.2.1 User Design

The UD stage produces a detailed system area model, an outline system design, and an implementation plan. End users participating in JAD workshops perform the analysis and design activities associated with this stage. This stage aims to analyze in detail business activities associated with the proposed system area. It also strives to analyze in detail the business data associated with the proposed system area. Additionally, it develops the system structure in terms of the automated and manual functions that will comprise the system and develop proposed screen layouts for the most important automated functions. The following illustrates this stage:

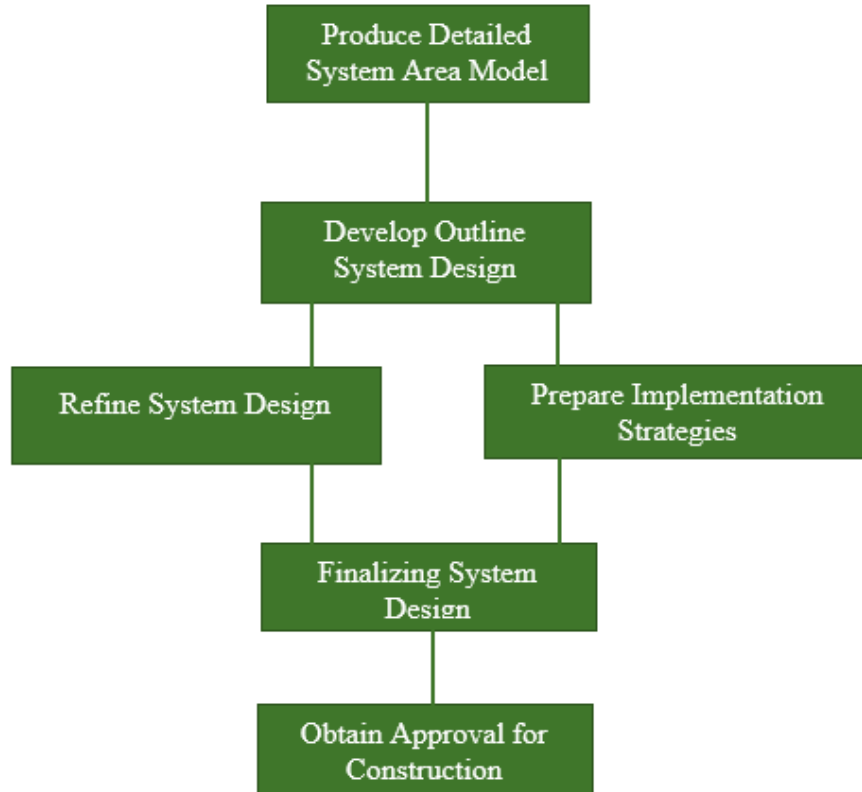


Figure 3: User Design

3.2.2 Rapid Construction

The design of the proposed system, initially described in the UD stage, is completed in the RC stage, and application software to implement that design is developed and tested. Activities to prepare for the transition of the system to production status are also performed.

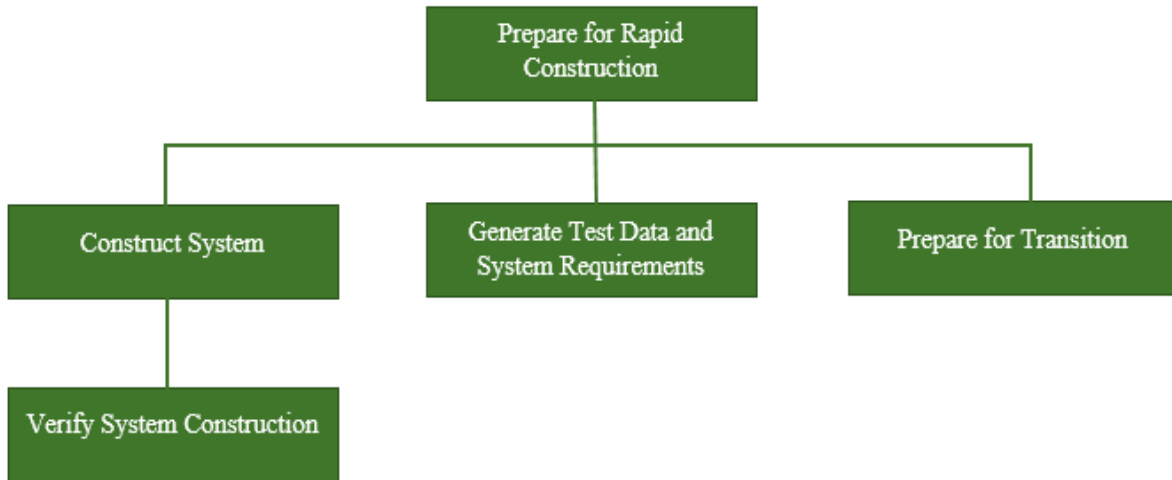


Figure 4: Rapid Construction

3.2.3 Transition

The system developed in the RC stage becomes operational in the TR stage. At this time, the developers prepare existing data for the new system and train users to operate the new application. They also provide support to resolve any problems that arise immediately after the application becomes operational.

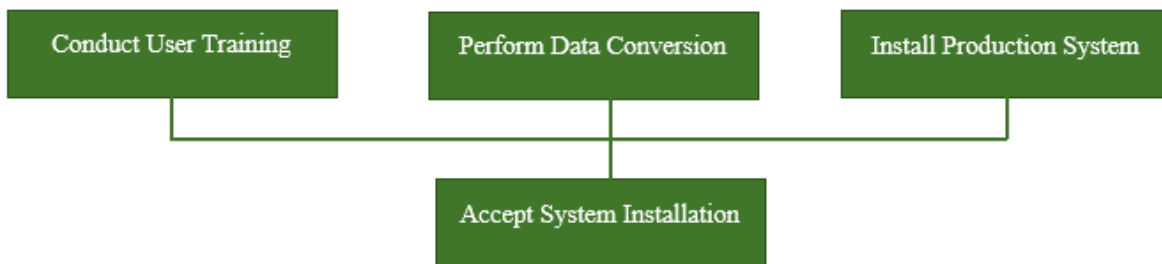


Figure 5: Transition

CHAPTER FOUR

SYSTEM DESIGN

4.1 INTRODUCTION

The most creative and challenging phase of the life cycle is system design. The term design describes a final system and the process by which it is developed. It refers to the technical specifications that will be applied in the implementation of the system. System design is the only way that user requirements can be translated into a complete system.

This chapter describes the system requirements, the process of analyzing a given system, defining problems, the system specifications, providing optimal solutions to a problem and the maintenance of the system.

4.2 SOFTWARE AND HARDWARE REQUIREMENTS

Software and hardware requirements describe the optimal environment within which the system should run. These requirements could describe the minimum requirements and the recommended requirements. The system was developed entirely using java8 and the following table shows the requirements this system requires to run on a computer.

Table 2:Software Requirements

Feature	Minimum Requirement	Recommended Requirement
Operating System	Windows Vista SP2 Intel based Mac OS X 10.8.3+ Any other Ubuntu/Arch based Linux Distro	Windows 8.x (Desktop) Windows 10 (8u51 and Above) Intel based Mac OS X 10.9+ Ubuntu Linux 15.10 (8u65 and above) / Manjaro linux 17.06
Database Management Tool	MySQL Workbench 5.0	MySQL Workbench 6.0+

Table 3: Hardware Requirements

Feature	Minimum Requirement	Recommended Requirement
Processor	Intel Pentium 2GHz	Intel Core i3 1.9GHz+
RAM	1GB	4GB+
HDD	80GB	250GB+

4.3 PROCESS

The process shows how data flows the system and eventually a timetable is generated. It also describes how various intended users interact with the system to generate or view the timetable.

The process can be illustrated using a flow chart and a use case diagram as shown below.

4.3.0 Use-Case Diagram

The system is used by three types of users; Admin, Lecturers and Students.

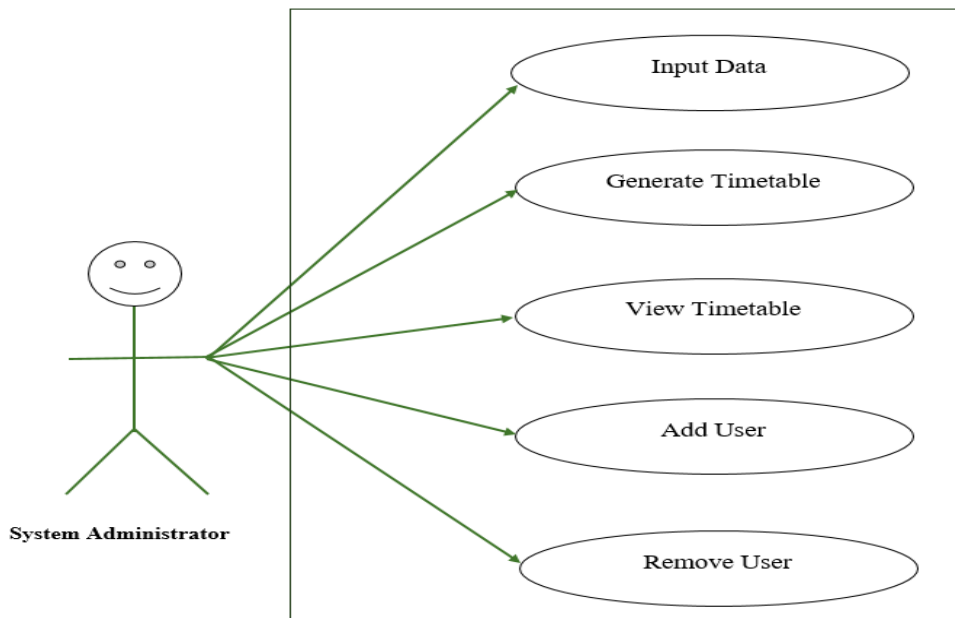


Figure 6: Administrator Use Case Diagram

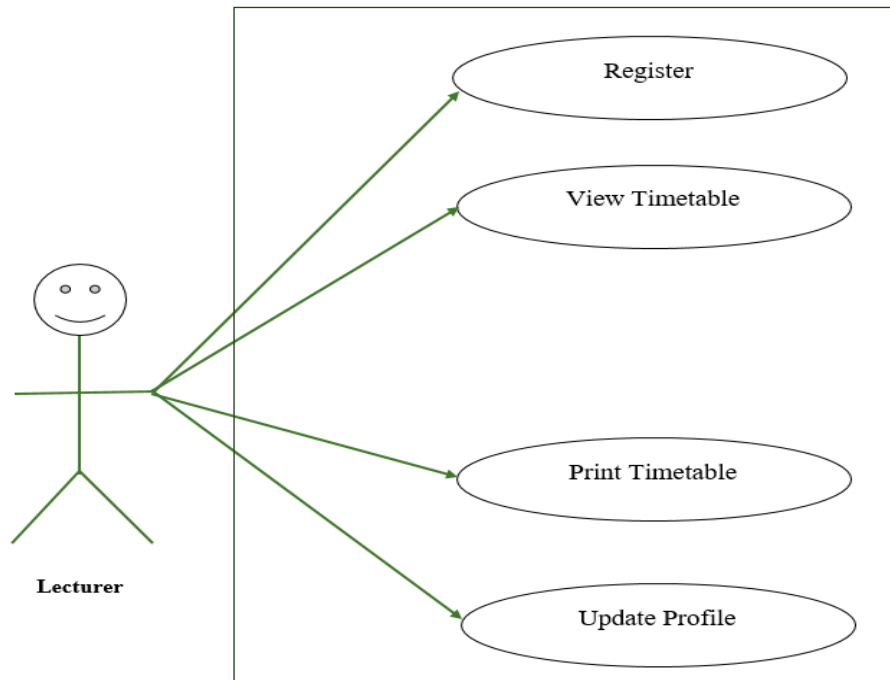


Figure 7: Lecturer Use Case Diagram

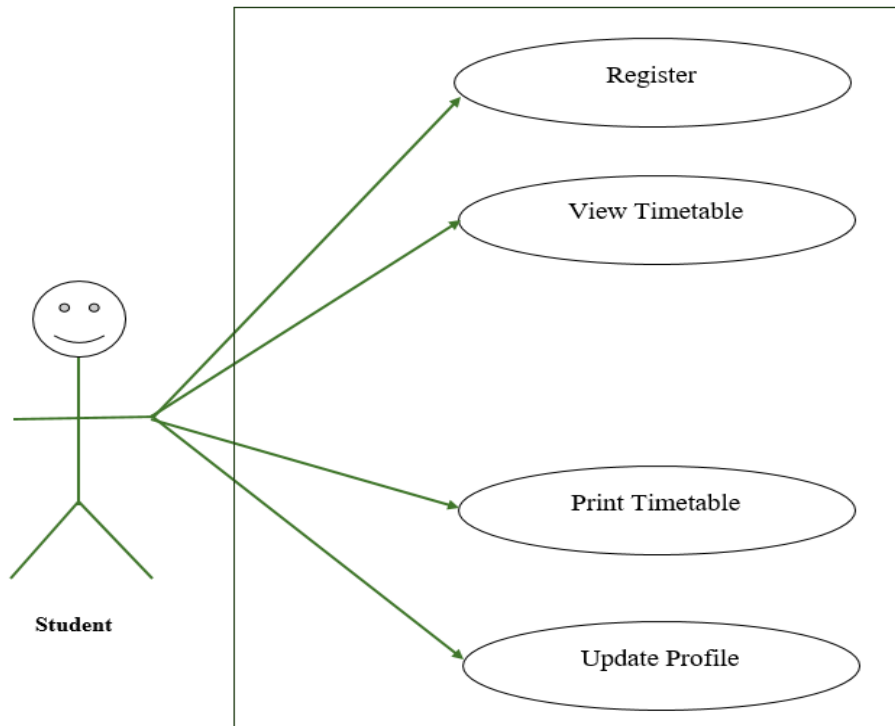


Figure 8: Student Use Case Diagram

4.3.1 Flow chart

The system flowchart is a visual representation of how data flows through the system, the processes it follows and the order of events.

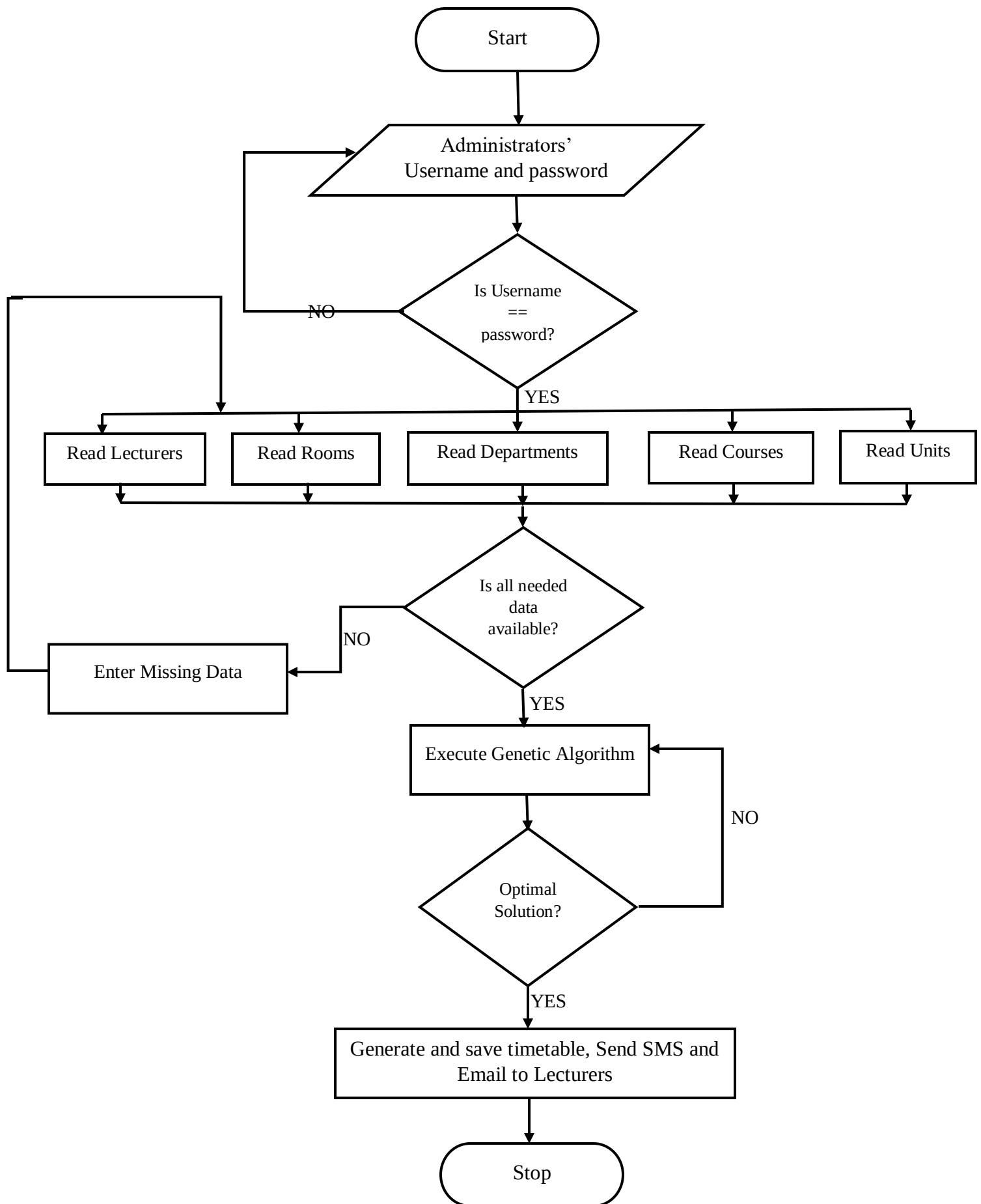


Figure 9: Flow chart showing the process of timetable generation

4.4 Input and Output

4.4.0 Context Diagram

Context diagram describes the system's interaction with its environment. The following is a use case diagram showing a high-level view of the system and what each type of user can do with the system and the system's feedback to each user.

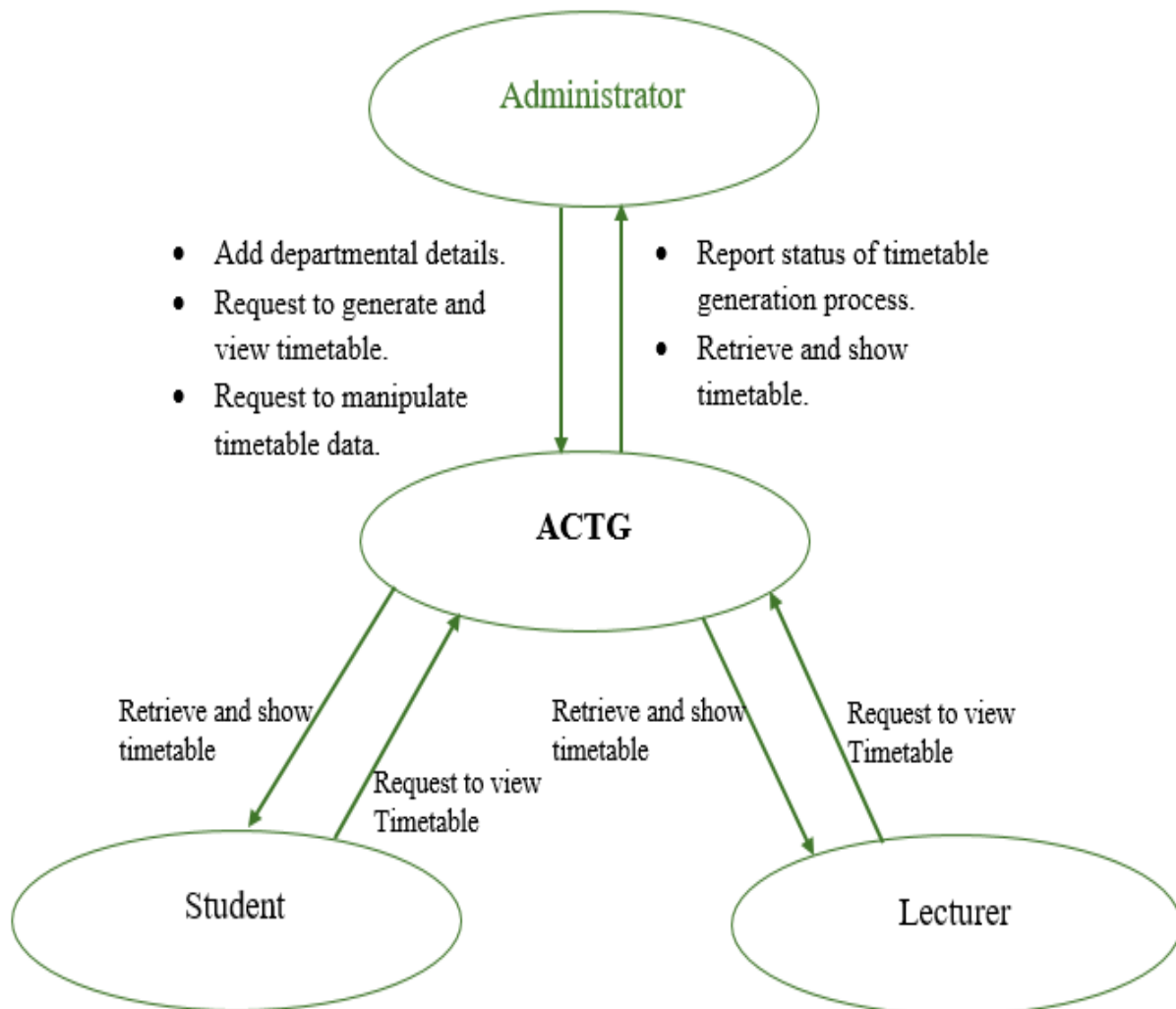


Figure 10: Context Diagram

4.5 User Interface

4.5.0 Login window

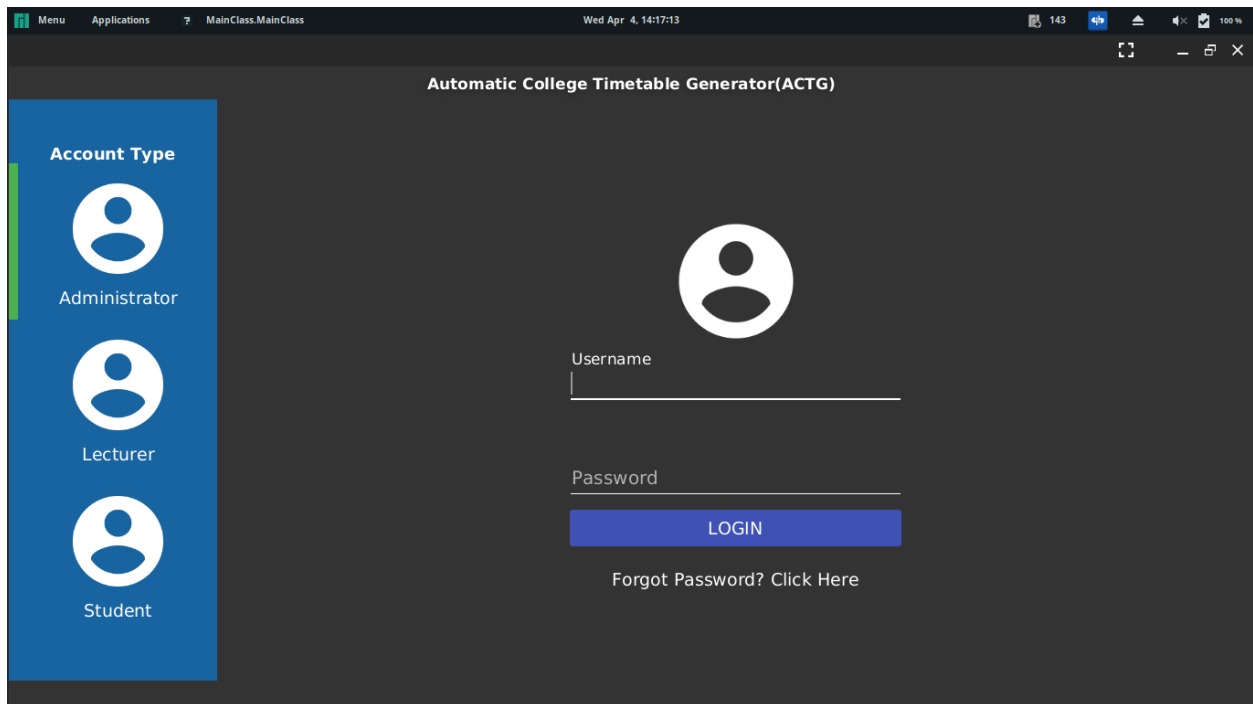


Figure 11: Administrators Login

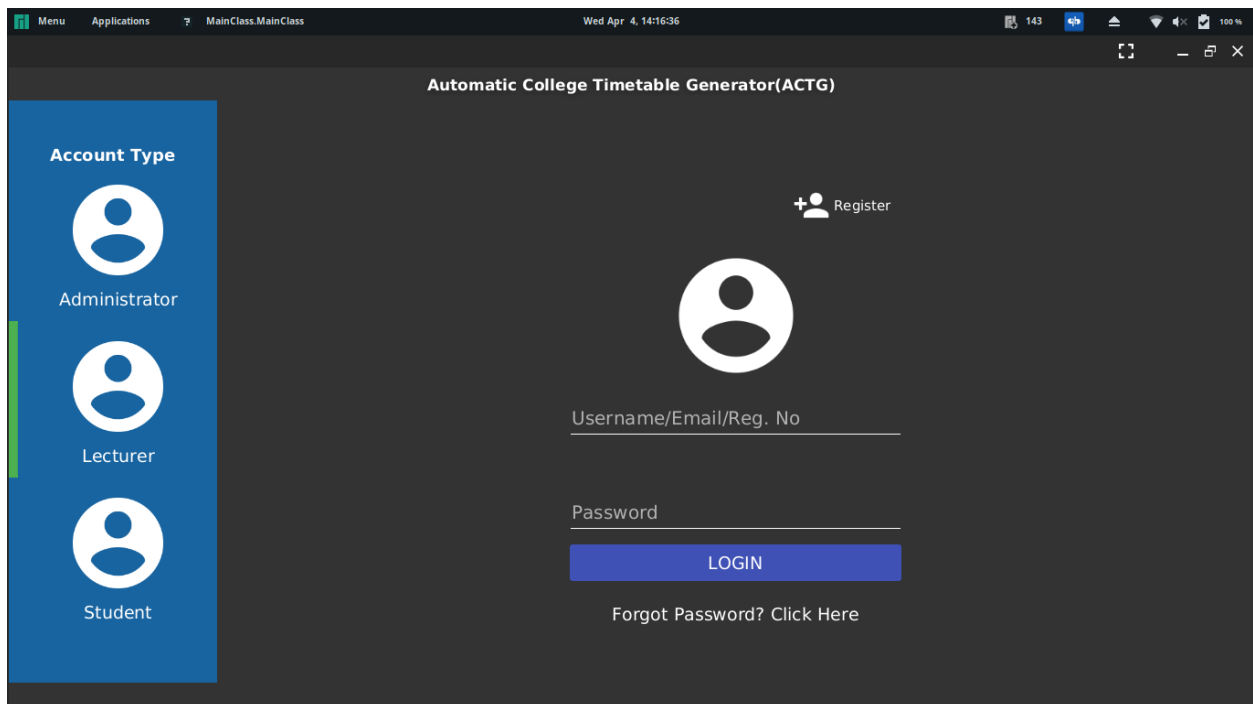


Figure 12: Lecturer or Student Login

This is the initial page that users interact with after starting the application. It ensures authentication and authorization of users.

The users submit their sign in credentials to the system and the system respond by checking whether there is a match between credentials entered and those in the database. It also consists of additional features such as password retrieval and new user registration (for students and lecturers only).

4.5.1 Main Window

4.5.1.0 Administrators Window

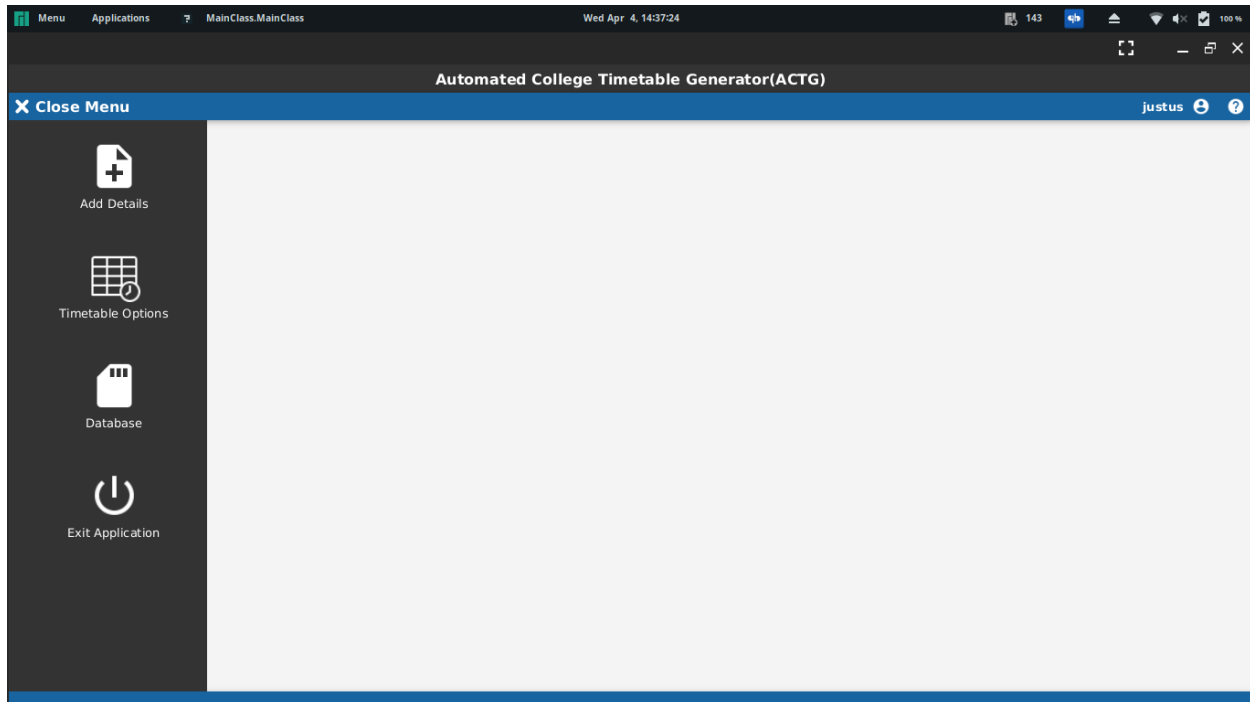


Figure 13:Administrators' Main Window

This is the panel for the system administrator. It has a menu on the right which contains various system controls such as data addition, timetable controls (viewing and generating controls), database viewing and manipulation.

4.5.1.1 Lecturers Window

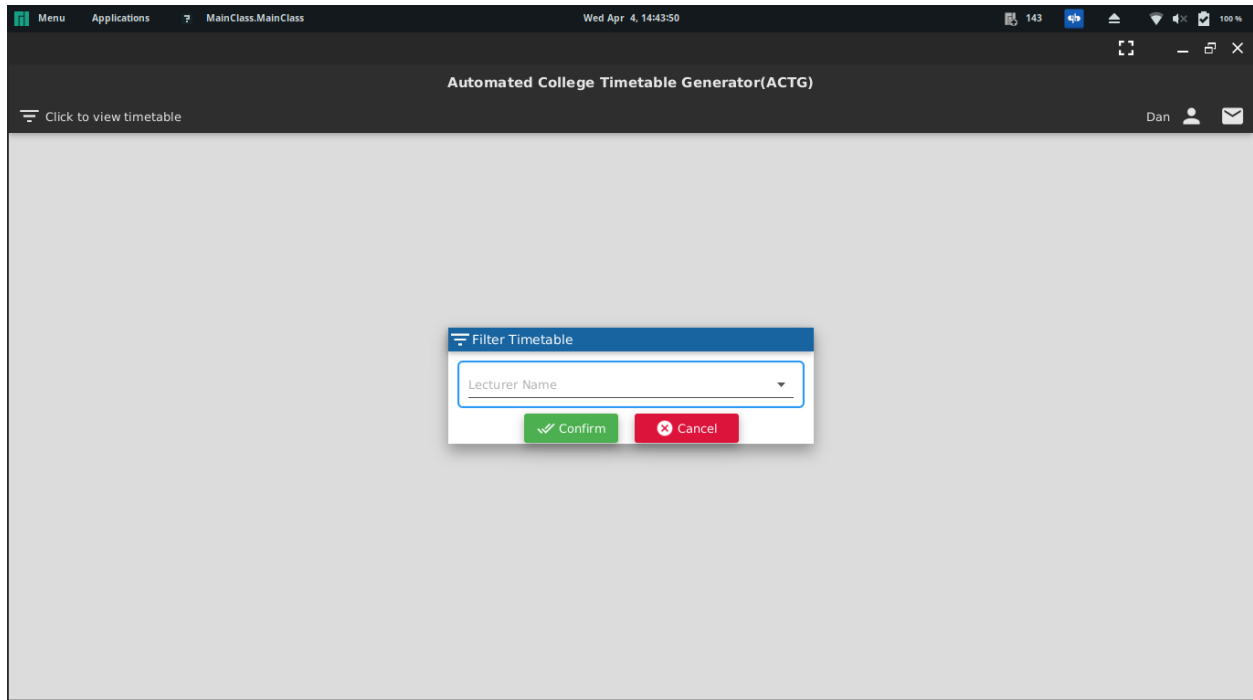


Figure 14: Lecturers' Main Window

Provide lecturers with the options to filter, view and print the timetable on their own. Also, they can be able to update their account profiles, logging out and composing messages to the system administrator in case of any complaints.

4.5.1.2 Students Window

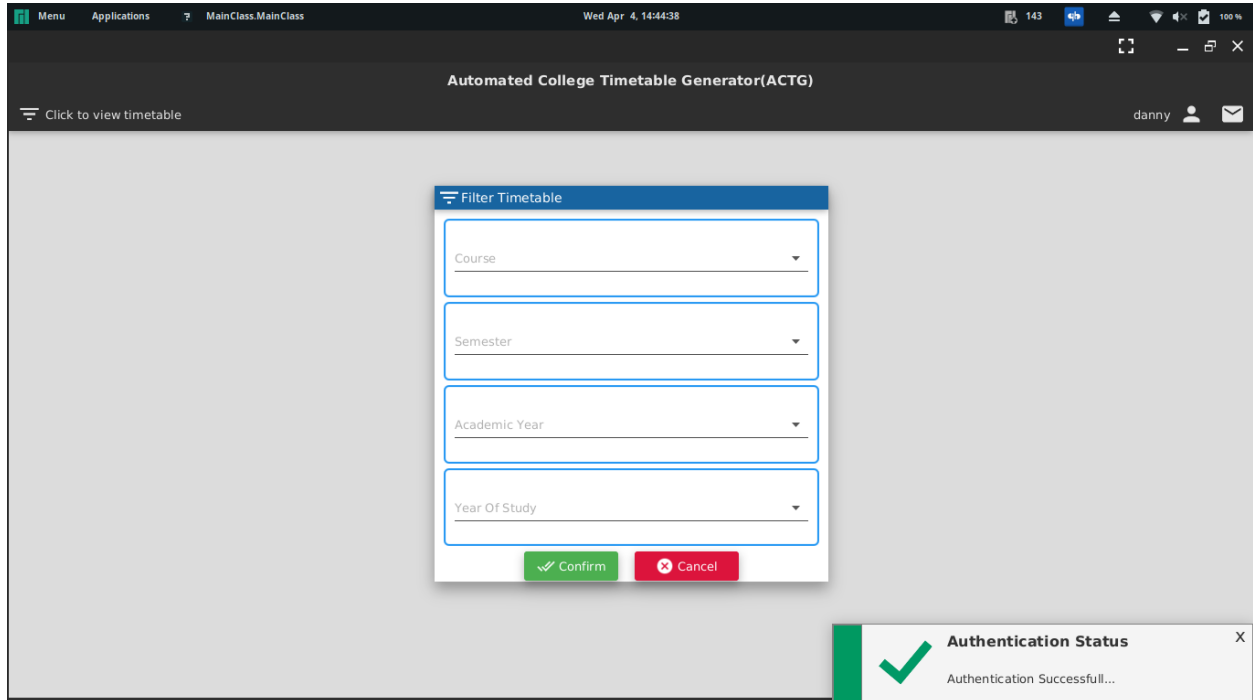


Figure 15: Students' Main Window

Similar features as the lecturers' window except that the filtering details required are more specific.

4.6 Database Design

The database schema used by ACTG is tabulated as follows.

4.6.0 course

Table 4: Course Table

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Course_Name	VARCHAR (45)	Null
Course_Code	VARCHAR (45)	Null
Number_Of_Students	INT (11)	Null
Number_Of_Units	INT (11)	Null
Department	VARCHAR (45)	Null
Semester	VARCHAR (45)	Null
Year_Of_Study	VARCHAR (45)	Null
Academic_Year	VARCHAR (45)	Null

4.6.1 department

Table 5:Table Department

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null

Name	VARCHAR (45)	Null
Abbreviation	VARCHAR (45)	Null

4.6.2 lecturer

Table 6: Table Lecturer

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Lecturer_Name	VARCHAR (45)	Null
Postal_Address	VARCHAR (45)	Null
Phone_Number	VARCHAR (45)	Null
Email	VARCHAR (45)	Null
Department	VARCHAR (45)	Null

4.6.3 meeting_time

Table 7: Table Meeting Time

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Time	VARCHAR (45)	Null
Available	VARCHAR (45)	Null

4.6.4 room

Table 8: Table Room

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Number	VARCHAR (45)	Null
Capacity	INT (11)	Null

4.6.5 timetable_data

Table 9: Table Timetable Data

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Department	VARCHAR (45)	Null
Course_Name	VARCHAR (45)	Null
Unit_Name	VARCHAR (45)	Null
Unit_Code	VARCHAR (45)	Null
Year_Of_Study	VARCHAR (45)	Null
Lecturer_Name	VARCHAR (45)	Null
Meeting_Time	VARCHAR (45)	Null
Room	VARCHAR (45)	Null
Semester	VARCHAR (45)	Null
Academic_Year	VARCHAR (45)	Null
Generation_Time	VARCHAR (45)	Null

4.6.6 tuition_details

Table 10: Table Tuition Details

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Academic_Year	VARCHAR (45)	Null
Year_Of_Study	VARCHAR (45)	Null
Semester	VARCHAR (45)	Null

4.6.7 unit

Table 11: Table Units

Column Name	Datatype	Constraint
ID	VARCHAR (45)	Primary Key, Not Null
Unit_Code	VARCHAR (45)	Null
Unit_Name	VARCHAR (45)	Null
Number_Of_Students	INT (11)	Null
Course_Name	VARCHAR (45)	Null
Lecturer_Name	VARCHAR (45)	Null

4.6.8 user_account

Table 12: Table User Account

Column Name	Datatype	Constraint
Reg_Number	VARCHAR (45)	Null
Full_Name	VARCHAR (45)	Null
Username	VARCHAR (45)	Null
Email	VARCHAR (45)	Primary Key, Not Null
Password_Hint	VARCHAR (45)	Null
Account_Type	VARCHAR (45)	Null
Photo_Upload	LOB	Null

CHAPTER FIVE

SYSTEM IMPLEMENTATION AND TESTING

5.0 INTRODUCTON

This chapter describes how the system will be implemented, tested and validated.

5.1 PROGRAMMING LANGUAGE

This system was implemented using Java SE 8, for the system's backend and JavaFX 8(GUI library for Java SE 8) for the user interface. SQL language was used to implement MySQL DBMS.

5.2 TOOLS USED

1. Netbeans IDE
2. IntelliJ IDEA
3. MySQL Workbech
4. Gmail Gateway
5. Africa's Talking SMS gateway
6. Manjaro Linux

5.2 IMPLEMENTATION STRATEGIES

According to [23], implementation strategies are aimed at allowing users to take over operation of a new system for use and evaluation. It involves training the users to handle the new system and plan for a smooth transition. The strategies used in system changeover include direct, phased, pilot and parallel running implementation. This system's implementation is a direct implementation strategy.

5.3 MODULE INTEGRATION AND TESTING

[24], describes this stage as where individual software modules are combined and tested. This occurs after the modules are tested separately, what is referred to as unit testing. For this system, the class with the code for the user interface was integrated with the class containing code for the timetable generation algorithm and to produce an integrated system ready for system testing.

5.3.0 SYSTEM TESTING

System testing is conducted based on the system's functional requirements. This system was tested to check how it accepted inputs, generated outputs and met its main objective. The following figures illustrate the test results.

5.3.0.0 System Inputs

Adding Tuition Details

The screenshot shows the 'Automated College Timetable Generator (ACTG)' application window. The title bar includes window controls and the text 'justus'. The main menu bar shows 'Open Menu' and 'Add Details - Add Tuition Details'. The background features four large buttons: 'Add Tuition Basics' (green), 'Add Department' (purple), 'Add Lecture Hall' (pink), and 'Add Time' (brown). A modal dialog box is open in the center, containing four text input fields labeled 'ID', 'Academic Year', 'Year Of Study', and 'Semester'. At the bottom of the dialog are 'Save' and 'Cancel' buttons.

Figure 16: Adding Tuition Details

Adding Rooms

The screenshot shows the 'Automated College Timetable Generator (ACTG)' application window. The title bar includes window controls and the text 'justus'. The main menu bar shows 'Open Menu' and 'Add Details - Add Lecture Hall'. The background features four large buttons: 'Add Tuition Basics' (green), 'Add Department' (purple), 'Add Lecture Hall' (pink), and 'Add Time' (brown). A modal dialog box is open in the center, containing three text input fields labeled 'ID', 'Number', and 'Capacity'. At the bottom of the dialog are 'Save' and 'Cancel' buttons.

Figure 17: Adding Rooms

Adding Departments

The screenshot shows the 'Automated College Timetable Generator (ACTG)' application window. The title bar includes window controls and the text 'Automated College Timetable Generator (ACTG)'. The menu bar shows 'Open Menu', 'Add Details - Add Department', and a user profile 'justus'. The main interface has four large buttons: 'Add Tuition Basics' (green), 'Add Department' (purple), 'Lecture Hall' (pink), and 'Add Time' (brown). A modal dialog box is open over the 'Add Department' button. The dialog has three input fields: 'ID', 'Name', and 'Abbreviation'. At the bottom of the dialog are 'Save' and 'Cancel' buttons.

Figure 18: Adding Departments

Adding Courses and Units

The screenshot shows the 'Automated College Timetable Generator (ACTG)' application window. The title bar includes window controls and the text 'Automated College Timetable Generator (ACTG)'. The menu bar shows 'Open Menu', 'Add Details - Add Course', and a user profile 'justus'. The main interface has four large buttons: 'Add Tuition Basics' (green), 'Add Department' (purple), 'Lecture Hall' (pink), and 'Add Time' (brown). A modal dialog box is open over the 'Add Department' button. The dialog has three input fields: 'ID', 'Name', and 'Abbreviation'. Below these is a section titled 'COURSE UNITS' with a table containing columns for 'Code', 'Number of Students', and 'Lecturer'. There is an 'Add Unit' button next to the table. At the bottom of the dialog are 'Save' and 'Cancel' buttons.

Figure 19: Adding Courses and Units

Adding Meeting Times

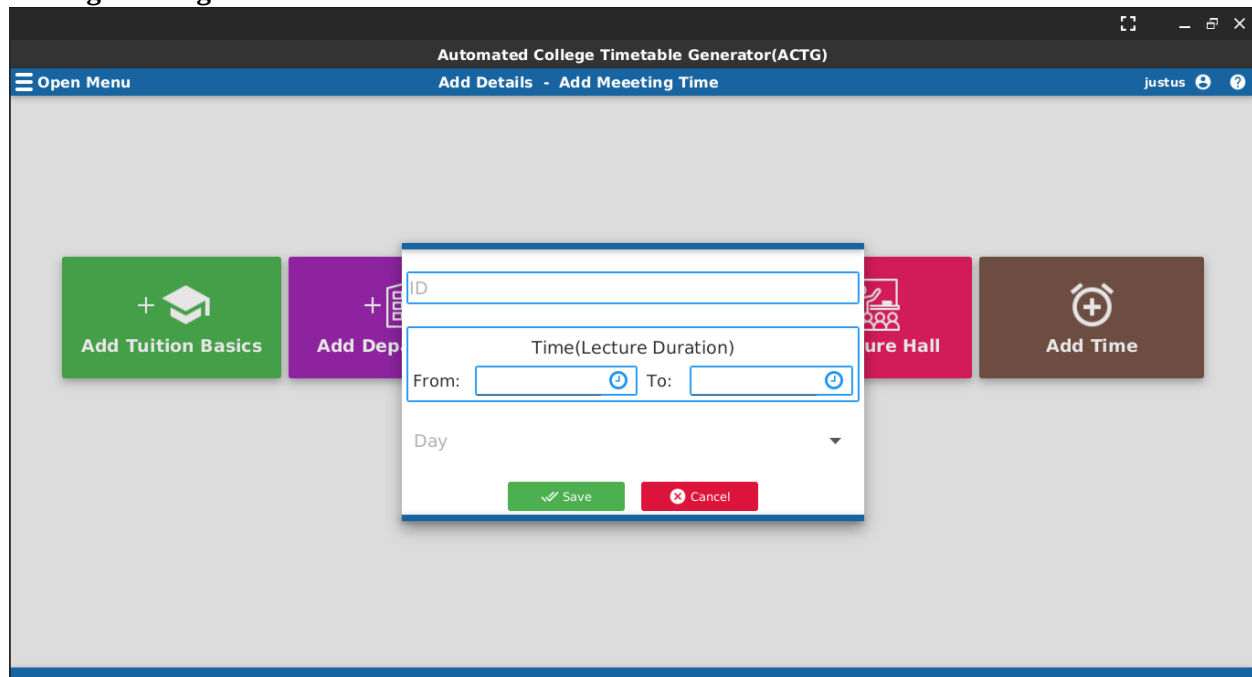


Figure 20: Adding Meeting Time

5.3.0.1 System Outputs

Generated Timetable

This tested that the system met its main objective. It clearly shows a timetable generated free of collisions in time and lecturers. This illustrates the main objective of the system; automatic generation of a timetable. On the screen, a timetable of a specific course being taken by students at a certain stage, such as fourth years, will be displayed. The system will also allow the user to print the timetable. The following pictures show generated timetable and how various users access it.

Entire Timetable View and Report

GENERATED ON :Wed 04/04/2018 at 01:46:53 PM EAT SEMESTER :Two ACADEMIC YEAR :4							
Lecturer Name							
Department	Course	Code	Unit Name	Year_Of_Study	Venue	Lecturer	Time
Pure & Applied Sciences	BSc Mathematics and ...	ICS 2365	Selected Topics in Com...	2017/2018	RM-5	John	THURSDAY (07:00 - 10:...
PAS	BSc Mathematics and ...	ICS 2364	Multimedia Systems	2017/2018	RM-2	John	FRIDAY (10:00 - 13:00)
PAS	BSc Mathematics and ...	ICS 2269	Computer Systems Proj...	2017/2018	RM-3	Daniel Kahiga Kamau	MONDAY (16:00 - 19:00)
Pure & Applied Sciences	BSc Mathematics and ...	SMA 2415	Complex Analysis II	2017/2018	RM-1	Daniel Kahiga Kamau	TUESDAY (13:00 - 16:00)
Pure & Applied Sciences	BSc Mathematics and ...	SMA 2463	Analytical Applied Math...	2017/2018	RM-2	John	WEDNESDAY (13:00 - 1...
Pure & Applied Sciences	Bsc Statistics	SMA 2413	Probability and Statisti...	2017/2018	RM-5	John	FRIDAY (13:00 - 16:00)
PAS	BSc Mathematics and ...	ICS 2365	Selected Topics in Com...	2017/2018	RM-7	John	WEDNESDAY (10:00 - 1...
PAS	Bsc Statistics	SMA 2452	Regression Analysis	2017/2018	RM-5	Daniel Kahiga Kamau	FRIDAY (10:00 - 13:00)

Print

TIMETABLE REPORT Entire Timetable							
SEMESTER: Two YEAR OF STUDY: 2017/2018 GENERATED ON: Wed 04/04/2018 at 01:46:53 PM							
Department	Course_Name	Unit_Name	Unit_Code	Lecturer_Name	Meeting_Time	Room	Academic_YR
Pure & Computer Science	BSc Mathematics and Computer Science	Selected Topics in Computer Science	ICS 2365	John	THURSDAY (07:00 - 10:00)	RM-5	4
PAS	BSc Mathematics and Computer Science	Multimedia Systems	ICS 2364	John	FRIDAY (10:00 - 13:00)	RM-2	4
PAS	BSc Mathematics and Computer Science	Computer Systems Project	ICS 2269	Daniel Kahiga Kamau	MONDAY (16:00 - 19:00)	RM-3	4
Pure & Computer Science	BSc Mathematics and Computer Science	Complex Analysis II	SMA 2415	Daniel Kahiga Kamau	TUESDAY (13:00 - 16:00)	RM-1	4
Pure & Computer Science	BSc Mathematics and Computer Science	Analytical Applied Mathematics II	SMA 2463	John	WEDNESDAY (13:00 - 15:00)	RM-2	4
Pure & Computer Science	Bsc Statistics	Probability and Statistics IV	SMA 2413	John	FRIDAY (13:00 - 16:00)	RM-5	4
PAS	BSc Mathematics and Computer Science	Selected Topics in Computer Science	ICS 2365	John	WEDNESDAY (10:00 - 13:00)	RM-7	4

Thu, 5 Apr 2018 08:34:51

Page 1 of 2

Page 1 of 2

Print

Report Generation Status

Timetable Report Generated Successfully

Figure 21: Entire Timetable view and Report

Automated College Timetable Generator(ACTG)

Dan

Lecturer Name :Daniel Kahiga Kamau

DEPARTMENT :PAS
SEMESTER :Two
YEAR OF STUDY:2017/2018

GENERATED ON :Wed 04/04/2018 at 01:46:54 PM EAT

Course	Unit		Venue	Time	Academic Year
	Code	Name			
BSc Mathematics and Computer...	ICS 2269	Computer Systems Project	RM-3	MONDAY (16:00 - 19:00)	4
BSc Mathematics and Computer...	SMA 2415	Complex Analysis II	RM-1	TUESDAY (13:00 - 16:00)	4
Bsc Statistics	SMA 2452	Regression Analysis	RM-5	FRIDAY (10:00 - 13:00)	4

Print

TIMETABLE REPORT

Filtered Timetable

LECTURER NAME:Daniel Kahiga Kamau
SEMESTER:Two
DEPARTMENT:PAS

YEAR OF STUDY:2017/2018
GENERATED ON:Wed 04/04/2018 at 01:46:53 PM EAT

Course_Name	Unit_Name	Unit_Code	Meeting_Time	Room	Academic_Year
BSc Mathematics and Computer Science	Computer Systems Project	ICS 2269	MONDAY (16:00 - 19:00)	RM-3	4
BSc Mathematics and Computer Science	Complex Analysis II	SMA 2415	TUESDAY (13:00 - 16:00)	RM-1	4
Bsc Statistics	Regression Analysis	SMA 2452	FRIDAY (10:00 - 13:00)	RM-5	4

Thu, 5 Apr 2018 08:32:53 +0300 Page 1 of 1

Dan

Lecturer Name :Daniel Kahiga Kamau

Time	Academic Year
MONDAY (16:00 - 19:00)	4
TUESDAY (13:00 - 16:00)	4
FRIDAY (10:00 - 13:00)	4

Report Generation Status

✓

Timetable Report Generated Successfully

Print

Automated College Timetable Generator(ACTG)

Click to view timetable

danny

DEPARTMENT :PAS

COURSE :BSc Mathematics and Computer Science

SEMESTER :Two

ACADEMIC YEAR :4

YEAR OF STUDY:2017/2018

GENERATED ON :Wed 04/04/2018 at 01:46:53 PM EAT

Unit		Venue	Lecturer	Time
Code	Name			
ICS 2365	Selected Topics in Computer Science	RM-5	John	THURSDAY (07:00 - 10:00)
ICS 2364	Multimedia Systems	RM-2	John	FRIDAY (10:00 - 13:00)
ICS 2269	Computer Systems Project	RM-3	Daniel Kahiga Kamau	MONDAY (16:00 - 19:00)
SMA 2415	Complex Analysis II	RM-1	Daniel Kahiga Kamau	TUESDAY (13:00 - 16:00)
SMA 2463	Analytical Applied Mathematics II	RM-2	John	WEDNESDAY (13:00 - 15:00)
ICS 2365	Selected Topics in Computer Science	RM-7	John	WEDNESDAY (10:00 - 13:00)

Print

Figure 23: Students' Timetable View and Report

5.3.1 INTEGRATING THE SYSTEM WITH DATABASE

This system automatically generates a timetable by fetching data stored in the database about the courses and units on offer during a certain semester, the available rooms and the lecturers available to teach the units on offer and runs a genetic algorithm to give a timetable as the output.

5.4 DATA MODEL

Data models define how the logical structure of a database is modeled. They introduce abstraction in a DBMS and define how data is related to each other and how it is processed and stored inside the system [25]. The figure below illustrates the data model of this system.

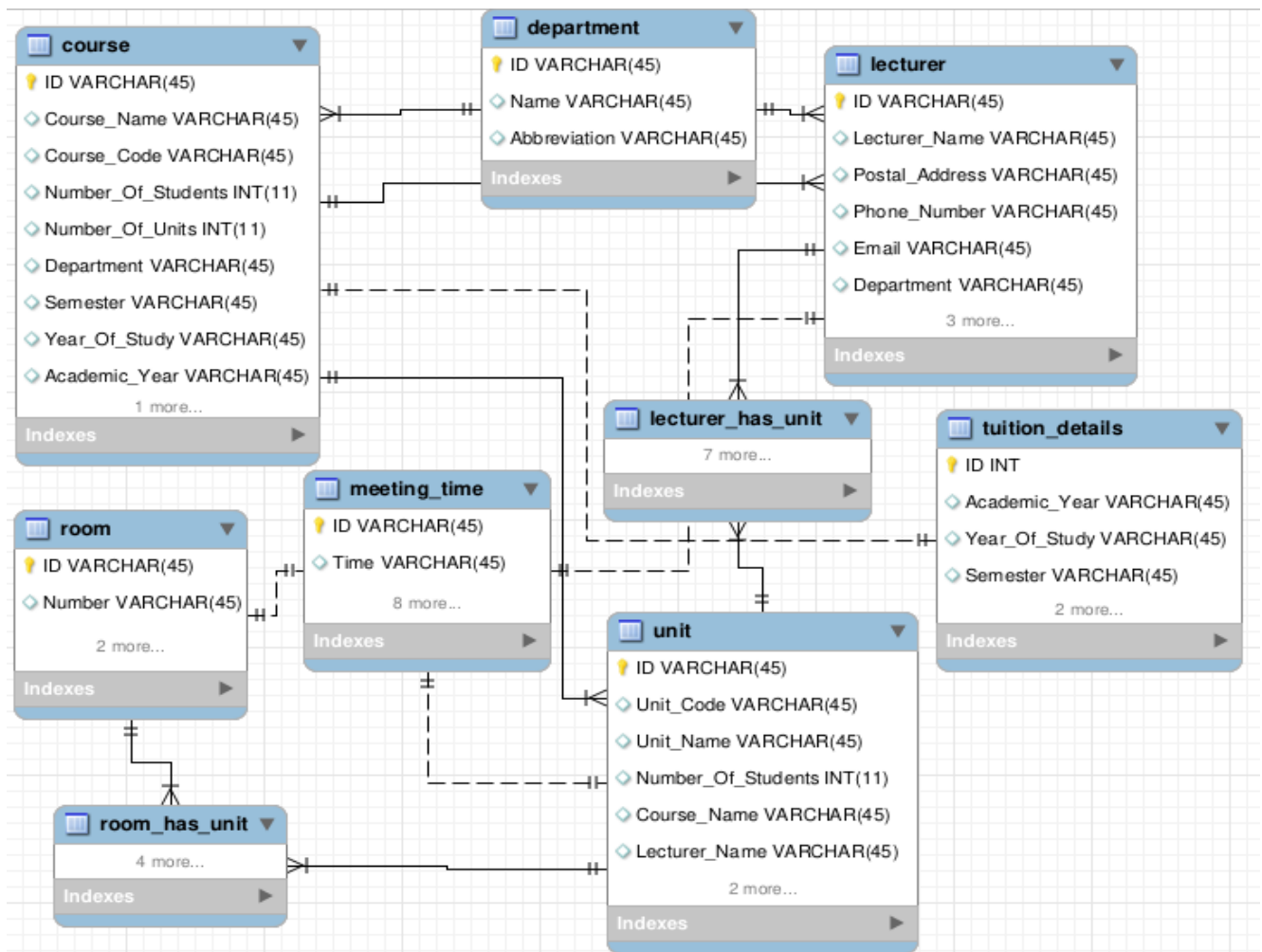


Figure 24: EER Diagram Showing Data Model

CHAPTER SIX

CONCLUSIONS AND RECOMMENDATIONS

6.0 INTRODUCTION

This chapter covers the overall summary of the system, its critical review, insights on future developments of the system and what the system has achieved along the challenges faced during development and implementation. A conclusion, recommendations and suggestions for further research in the future is made.

6.1 OVERALL SUMMARY

This system's main objective is to automatically generate a college timetable. This was meant to optimize the process of timetabling by reducing the time and energy used to manually prepare timetables at Colleges. A genetic algorithm was implemented to enable the system to meet its specific objectives. The system can fetch data from the database to populate functions in the algorithm, subsequently allocating classrooms to class sessions efficiently, assigning class sessions to available time periods and optimally distributing units on offer to available lecturers. By performing these functions, the system automatically generates a timetable in a few seconds, saving on time. Rapid application development was used to develop this system to utilize the available time for its completion. The implementation of the algorithm required Java programming language because it is inherent in the language.

6.2 FUTURE DEVELOPMENTS

One of the important aspects of a system is its scalability. This system is no exception and it has areas that could be considered for further development. To begin with, the system could generate examination timetable as well.

6.3 ACHIEVEMENTS MADE

ACTG allows faculty staff responsible for coordinating the preparation of timetables to enter data about lecturers, rooms, departments, courses and units into the system, and view the availability of these resources to make appropriate decisions about the timetabling process. Most importantly,

the system can generate a timetable and make available the output in electronic form or in printed form.

6.4 CHALLENGES FACED

The researcher incurred extra costs that had not been anticipated to fully understand the concept of genetic algorithms inherent in the java programming language. This was further escalated by the high costs of internet connectivity.

6.5 RECOMMENDATIONS

Owing to the dynamism of college admissions and academic programs, it is recommended that the institution maintains a working relationship with the developer of this system to allow for continuous updating of the system to address changing policies and any vulnerabilities that could be identified with continued use of the system.

6.6 SUGGESTIONS FOR FUTURE WORK

While this system achieves efficiency in timetable generation, future work could accommodate more soft constraints to allow for student and lecturer preferences such as which time periods are best for certain units. Further research could also be conducted on how such a system could be integrated with an exam timetabling system.

6.7 CONCLUSION

Following the completion of this project, it is evident that the system produced the desired results and met its objectives by automatically generating timetables. Its significance to college education, can also be verified because it saves faculty staff time and allows lecturers and students to easily access the timetables by themselves.

BIBIOGRAPHY

References

1. Kuhn HW, 1955. The Hungarian method for the assignment problem. Naval res Logist quart.
2. Haynes JG, 1959. The conference scheduling problem. Operations research society of America 16th annual meeting Pasadena California.
3. Gotlieb C C, 1963. The construction of class-teacher timetables. Proceedings of IFIP congress 1962 Amsterdam.
4. Dowsland, 1990. A Timetabling Problem in Which Clashes Are Inevitable. Journal-of-the-Operational-Research-Society. 5079
5. Abramson, D. 1991. Constructing School Timetables Using Simulated Annealing: Sequential and Parallel Algorithms Management-Science.
6. Abramson, D., & Abela, J. (1991). A parallel genetic algorithm for the solving the school timetabling problem. In *Proceedings of the fifteenth Australian conference: division of information technology, C.S.I.R.O.* (pp. 1–11).
7. Moschopoulos, C. N., Alexakos, C. E., Dosi, C., Beligiannis, G. N., & Likothanassis, S. D. (2009). A user friendly evolutionary tool for high-school timetabling. *Studies in Computational Intelligence*, 166, 149–162.
8. Burke, E. K., Elliman, D. G., and Weare, R. F 1994-1995. The automation of the timetabling process in higher education Journal of Educational Technology Systems
9. Mitchell, M. (1996). *An introduction to genetic algorithms*. Cambridge, Mass: MIT Press.
10. Michalewicz, Z. (1992). *Genetic algorithms + data structures =: Evolution programs*. Berlin: Springer-Verlag.

11. Gwiazda, T. D. (2006). *Crossover for single-objective numerical optimization problems*.
Łomianki: TOMASZGWIAZDA E-BOOKS.
12. Kingston, J. H. (2007). Hierarchical timetable construction. In E. Burke & H. Rudova (Eds.), *Lecture notes in computer science: Vol. 3867. PATAT VI* (pp. 294–307).
13. Kingston, J. H. (2010). Solving the general high school timetabling problem. In *Proceedings of the 8th international conference on the practice and theory of automated timetabling (PATAT 2010)* (pp. 517–518).
14. Srndic, N., Dervisevic, M., Pandzo, E., & Konjicija, S. (2009). The application of a parallel genetic algorithm to timetabling of elementary school classes: a coarse grained approach. In *Proceedings of ICAT 2009–2009 22nd international symposium on information, communication and automation technologies* (pp.1–5). New York: IEEE Press.
15. Gudes-E, Kuflik-T, and Meisels-A, 1990. On resource allocation by an expert system
Engineering-Applications-of-Artificial-Intelligence.
16. Solotorevsky-G, Gudes-E, and Meisels-A, 1991. Specifying resource allocation and timetabling problems using a rule-based language. *Proceedings. The Fifth Israel Conference on Computer Systems and Software Engineering* (Cat. No. 91TH0349-1). IEEE Comput. Soc. Press, Los Alamitos, CA, USA; 1991; xiv+190 pp.
17. Reeves C, 1993. *Modern Heuristic Techniques for Combinatorial Problems*, Blackwell Scientific Publications, Oxford.
18. Hertz-A, 1991. Tabu search for large scale timetabling problems. *European Journal-of-Operational-Research*.
19. Brailsford, S. C., Potts, C. N., and Smith, B. M. 1999. Constraint satisfaction problems: Algorithms and applications. *European Journal of Operational Research* 119.

20. Deris-S, Omatu-S, and Ohta-H, 1995. Course scheduling and timetabling by constraint-based reasoning. Bulletin-of-University-of-Osaka-Prefecture, -Series A.
21. Frangouli-H, Harmandas-V, and Stamatopoulos-P, 1995. UTSE: construction of optimum timetables for university courses-a CLP based approach. Practical Application of PROLOG. The Proceedings of the Third International Conference on the Practical Application of Prolog. Alinmead Software, Blackpool, UK; 1995; 719 pp.
22. Deris S, Omatu S Hiroshi Ohta, Saada, P., 1999. Incorporating constraint propagation in genetic algorithm for university timetable planning. Engineering Applications of Artificial Intelligence (12) pp. 241-253.
23. Rainer, R. K., & Cegielski, C. G. (2011). *Introduction to information systems*. Hoboken, N.J: John Wiley & Sons.
24. Jawadekar, W. S. (2013). *Management information systems: Text and cases : a global digital enterprise perspective*. New Delhi: McGraw Hill Education (India)
25. Hirschheim, R. A., Klein, H.-K., & Lyytinen, K. (1995). *Information systems development and data modeling: Conceptual and philosophical foundations*. Cambridge: Cambridge University Press

APPENDICES

Schedule

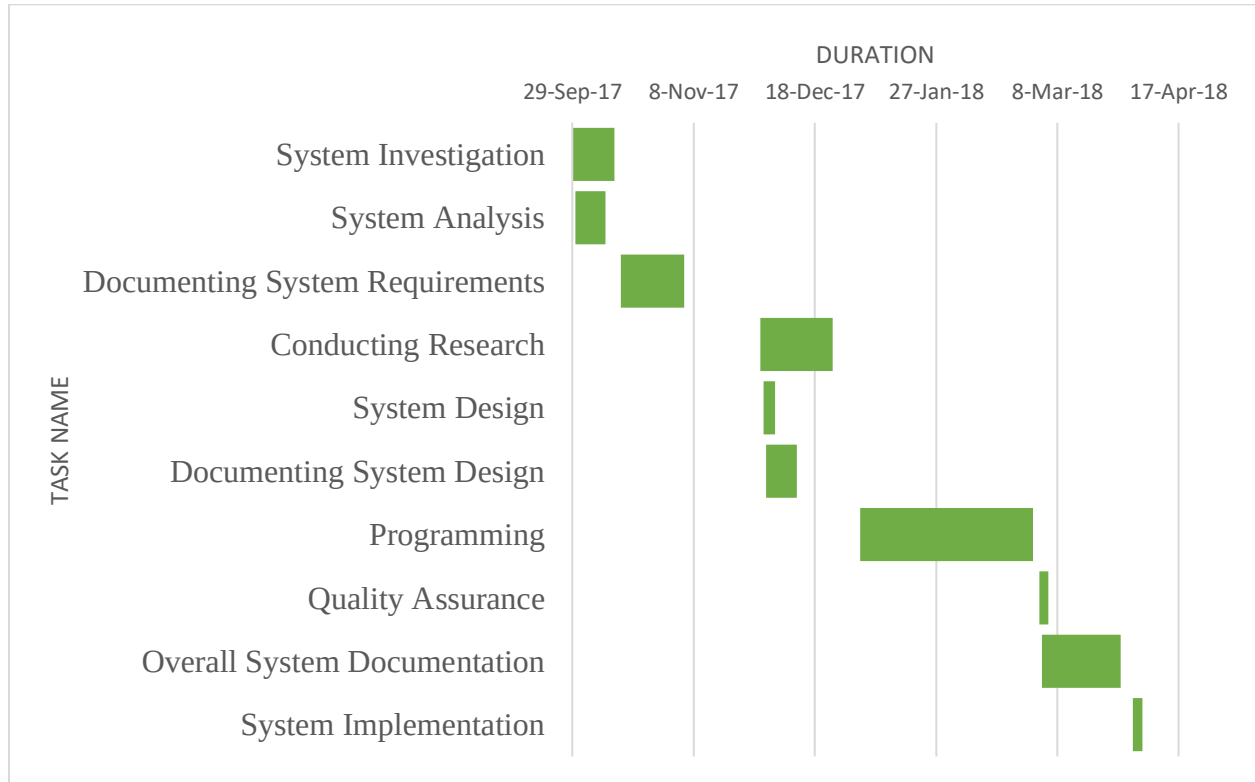


Figure 25: Schedule Gantt Chart

Budget

Table 13: System Budget

ITEM	COST (\$)
Laptop	349.36
SMS credit block	7.8
Stationery	4.85
Research materials(Internet)	19.41
External Drives (Flash Drive)	11.69